



JPMethod Polynomials

Algebraic Expressions

Polynomials & Quadratic Equations

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Algebraic Expressions - Polynomials & Quadratic Equations Instructions

This workbook teaches polynomial manipulation and quadratic equation solving through progressive exercises.

- Start with polynomial multiplication
- Learn multiplication using algebraic formulas
- Master factorisation techniques
- Work with square roots and quadratic equations
- Explore graphs of quadratic functions
- Apply the Pythagorean theorem

Regular practice with polynomials builds strong algebraic foundations.

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 2a(a + 4) =$$

$$(2) \quad 2a(a + 9) =$$

$$(3) \quad (x + 1)(x + 1) =$$

$$(4) \quad 3a(a - 9) =$$

$$(5) \quad 5a(a + 1) =$$

$$(6) \quad (y + 3)(y - 2) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 3x(x - 7) =$$

$$(2) \quad 4x(x - 1) =$$

$$(3) \quad (y - 1)(y + 2) =$$

$$(4) \quad (y - 4)(y + 2) =$$

$$(5) \quad 4a(a - 7) =$$

$$(6) \quad (x - 2)(x + 5) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 2x(x - 3) =$$

$$(2) \quad 5y(y - 8) =$$

$$(3) \quad 2a(a - 5) =$$

$$(4) \quad 4y(y + 8) =$$

$$(5) \quad (x - 1)(x + 6) =$$

$$(6) \quad 4x(x + 9) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 2y(y - 6) =$$

$$(2) \quad (x + 4)(x - 5) =$$

$$(3) \quad 4a(a + 6) =$$

$$(4) \quad 2a(a + 5) =$$

$$(5) \quad 2y(y - 8) =$$

$$(6) \quad 4y(y + 5) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad (y - 1)(y - 1) =$$

$$(2) \quad 2x(x + 2) =$$

$$(3) \quad (x - 2)(x + 2) =$$

$$(4) \quad (y + 2)(y - 5) =$$

$$(5) \quad (y - 4)(y + 5) =$$

$$(6) \quad 3x(x + 1) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 3x(x + 8) =$$

$$(2) \quad (x + 3)(x + 4) =$$

$$(3) \quad (x + 1)(x + 6) =$$

$$(4) \quad 3y(y + 8) =$$

$$(5) \quad (x + 2)(x - 4) =$$

$$(6) \quad (y - 3)(y + 4) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 4x(x + 9) =$$

$$(2) \quad (x - 1)(x + 5) =$$

$$(3) \quad 2x(x + 2) =$$

$$(4) \quad (x + 2)(x + 5) =$$

$$(5) \quad (y - 3)(y + 2) =$$

$$(6) \quad (y - 2)(y - 6) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 2x(x + 1) =$$

$$(2) \quad 2y(y + 9) =$$

$$(3) \quad (y + 2)(y - 5) =$$

$$(4) \quad 5y(y + 2) =$$

$$(5) \quad (y - 4)(y + 4) =$$

$$(6) \quad 2y(y + 6) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad (x + 1)(x + 4) =$$

$$(2) \quad 4x(x + 2) =$$

$$(3) \quad (y - 2)(y + 4) =$$

$$(4) \quad (x + 2)(x + 3) =$$

$$(5) \quad 4y(y - 4) =$$

$$(6) \quad 4a(a + 5) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad (x - 2)(x - 6) =$$

$$(2) \quad 4x(x - 2) =$$

$$(3) \quad 5x(x - 8) =$$

$$(4) \quad (x + 2)(x - 1) =$$

$$(5) \quad (y - 1)(y - 3) =$$

$$(6) \quad (x + 1)(x - 3) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 2a(a - 6) =$$

$$(2) \quad (x + 4)(x - 5) =$$

$$(3) \quad 5x(x + 9) =$$

$$(4) \quad (y - 1)(y - 6) =$$

$$(5) \quad 4a(a - 7) =$$

$$(6) \quad (y - 2)(y - 2) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 5y(y - 1) =$$

$$(2) \quad 4x(x - 8) =$$

$$(3) \quad 2x(x + 3) =$$

$$(4) \quad (x + 4)(x - 2) =$$

$$(5) \quad (x + 3)(x + 2) =$$

$$(6) \quad (y + 2)(y + 2) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 4y(y - 1) =$$

$$(2) \quad (x - 4)(x - 5) =$$

$$(3) \quad (y - 2)(y + 5) =$$

$$(4) \quad 4a(a + 6) =$$

$$(5) \quad 4x(x + 3) =$$

$$(6) \quad 3y(y - 2) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 4y(y + 9) =$$

$$(2) \quad 3x(x + 3) =$$

$$(3) \quad 5y(y - 9) =$$

$$(4) \quad (x - 2)(x + 4) =$$

$$(5) \quad (y - 1)(y - 3) =$$

$$(6) \quad (x - 4)(x - 2) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad (x - 4)(x + 3) =$$

$$(2) \quad (x + 1)(x + 4) =$$

$$(3) \quad (x + 4)(x - 2) =$$

$$(4) \quad (y - 2)(y - 4) =$$

$$(5) \quad (y + 4)(y - 3) =$$

$$(6) \quad 2y(y + 2) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad (x - 4)(x + 4) =$$

$$(2) \quad 2x(x + 4) =$$

$$(3) \quad (x - 2)(x + 1) =$$

$$(4) \quad (y - 4)(y - 5) =$$

$$(5) \quad 5y(y - 4) =$$

$$(6) \quad (x + 3)(x - 4) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 3x(x + 3) =$$

$$(2) \quad 5a(a - 4) =$$

$$(3) \quad (y + 2)(y + 5) =$$

$$(4) \quad (x - 1)(x - 3) =$$

$$(5) \quad 2a(a + 4) =$$

$$(6) \quad (x - 1)(x - 3) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad (y - 3)(y + 3) =$$

$$(2) \quad 2x(x - 7) =$$

$$(3) \quad 5a(a - 8) =$$

$$(4) \quad 2y(y + 7) =$$

$$(5) \quad (y - 4)(y - 5) =$$

$$(6) \quad (y - 4)(y - 3) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 2a(a + 6) =$$

$$(2) \quad 3x(x - 8) =$$

$$(3) \quad (x + 2)(x - 3) =$$

$$(4) \quad 4x(x + 9) =$$

$$(5) \quad (x - 1)(x + 5) =$$

$$(6) \quad (x - 1)(x - 5) =$$

Section 1. Multiplication of Polynomials I

Multiply the polynomials and simplify your answer.

$$(1) \quad 4x(x + 8) =$$

$$(2) \quad (y + 4)(y - 6) =$$

$$(3) \quad 4a(a - 8) =$$

$$(4) \quad (y + 3)(y - 2) =$$

$$(5) \quad (x - 4)(x - 5) =$$

$$(6) \quad 5y(y + 8) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad (y + 3)(y - 2) =$$

$$(2) \quad (x - 1)(x + 4) =$$

$$(3) \quad 3x(x + 5) =$$

$$(4) \quad 2a(a - 4) =$$

$$(5) \quad (y - 3)(y - 5) =$$

$$(6) \quad 2x(x - 4) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad 5a(a - 7) =$$

$$(2) \quad (y + 1)(y - 5) =$$

$$(3) \quad (y + 1)(y + 1) =$$

$$(4) \quad (x - 2)(x - 4) =$$

$$(5) \quad 5a(a - 2) =$$

$$(6) \quad (y + 3)(y + 3) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad (y - 4)(y + 3) =$$

$$(2) \quad (x - 3)(x + 3) =$$

$$(3) \quad (x + 4)(x + 4) =$$

$$(4) \quad (y + 4)(y - 1) =$$

$$(5) \quad 4y(y + 8) =$$

$$(6) \quad 3a(a + 5) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad 3x(x + 9) =$$

$$(2) \quad 2y(y - 8) =$$

$$(3) \quad 3y(y - 5) =$$

$$(4) \quad 3y(y - 4) =$$

$$(5) \quad 3a(a + 5) =$$

$$(6) \quad 2x(x - 5) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad (x - 2)(x + 1) =$$

$$(2) \quad 5a(a - 9) =$$

$$(3) \quad (y - 4)(y + 2) =$$

$$(4) \quad (x - 2)(x - 1) =$$

$$(5) \quad (y - 1)(y - 3) =$$

$$(6) \quad 5a(a - 6) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad (y + 4)(y + 5) =$$

$$(2) \quad 4x(x - 2) =$$

$$(3) \quad 2y(y + 5) =$$

$$(4) \quad 4x(x - 6) =$$

$$(5) \quad (x - 2)(x + 5) =$$

$$(6) \quad 3x(x + 7) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad (x - 2)(x - 4) =$$

$$(2) \quad (x + 4)(x + 3) =$$

$$(3) \quad (y - 3)(y - 6) =$$

$$(4) \quad (y - 2)(y + 4) =$$

$$(5) \quad 5a(a - 5) =$$

$$(6) \quad 5y(y - 1) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad (y + 3)(y - 2) =$$

$$(2) \quad 5x(x - 8) =$$

$$(3) \quad (x + 4)(x - 1) =$$

$$(4) \quad 5x(x + 8) =$$

$$(5) \quad 5a(a - 2) =$$

$$(6) \quad (y + 4)(y - 6) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad (x + 3)(x - 2) =$$

$$(2) \quad 2a(a - 1) =$$

$$(3) \quad (x + 3)(x + 6) =$$

$$(4) \quad (y + 4)(y + 2) =$$

$$(5) \quad (y - 2)(y - 3) =$$

$$(6) \quad (x + 4)(x + 1) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad (y + 4)(y - 5) =$$

$$(2) \quad 5a(a + 5) =$$

$$(3) \quad (x + 4)(x - 4) =$$

$$(4) \quad 4a(a + 6) =$$

$$(5) \quad (y - 2)(y + 1) =$$

$$(6) \quad 2x(x + 4) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad 2a(a + 4) =$$

$$(2) \quad (x + 1)(x + 3) =$$

$$(3) \quad (x + 1)(x + 6) =$$

$$(4) \quad 4x(x + 6) =$$

$$(5) \quad 4x(x + 7) =$$

$$(6) \quad 5y(y + 9) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad (y + 1)(y - 4) =$$

$$(2) \quad 5y(y - 5) =$$

$$(3) \quad (x + 2)(x - 4) =$$

$$(4) \quad 2a(a - 5) =$$

$$(5) \quad 3x(x - 5) =$$

$$(6) \quad 2x(x - 3) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad (x - 1)(x - 6) =$$

$$(2) \quad 2y(y + 7) =$$

$$(3) \quad (y - 1)(y + 1) =$$

$$(4) \quad 3y(y - 9) =$$

$$(5) \quad (y - 4)(y + 5) =$$

$$(6) \quad 3y(y - 4) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad (x - 1)(x + 3) =$$

$$(2) \quad (x + 4)(x - 5) =$$

$$(3) \quad 3y(y - 5) =$$

$$(4) \quad (x - 2)(x + 2) =$$

$$(5) \quad 2x(x + 9) =$$

$$(6) \quad (y + 4)(y - 3) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad (y + 1)(y + 1) =$$

$$(2) \quad 2y(y - 7) =$$

$$(3) \quad (y + 3)(y - 2) =$$

$$(4) \quad (x - 3)(x - 6) =$$

$$(5) \quad 3y(y + 1) =$$

$$(6) \quad 4x(x - 5) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad (y + 4)(y - 4) =$$

$$(2) \quad (y + 3)(y - 3) =$$

$$(3) \quad 2x(x + 7) =$$

$$(4) \quad (x - 1)(x + 5) =$$

$$(5) \quad (y + 4)(y - 6) =$$

$$(6) \quad (y + 1)(y + 5) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad (x - 1)(x + 4) =$$

$$(2) \quad 5a(a - 4) =$$

$$(3) \quad 5a(a + 7) =$$

$$(4) \quad (x - 3)(x - 5) =$$

$$(5) \quad (y + 1)(y - 4) =$$

$$(6) \quad 5a(a - 2) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad 5x(x - 8) =$$

$$(2) \quad (x + 3)(x - 5) =$$

$$(3) \quad (y - 1)(y - 1) =$$

$$(4) \quad 2a(a - 9) =$$

$$(5) \quad (x + 4)(x - 5) =$$

$$(6) \quad (x + 1)(x - 4) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad 3x(x - 8) =$$

$$(2) \quad 4y(y - 7) =$$

$$(3) \quad (x - 3)(x + 1) =$$

$$(4) \quad (y + 3)(y - 6) =$$

$$(5) \quad 3y(y + 7) =$$

$$(6) \quad 4a(a + 6) =$$

Section 2. Multiplication of Polynomials II

Multiply the polynomials and simplify your answer.

$$(1) \quad 3y(y + 7) =$$

$$(2) \quad (y - 3)(y + 2) =$$

$$(3) \quad 2a(a - 4) =$$

$$(4) \quad (x + 2)(x - 1) =$$

$$(5) \quad 3a(a + 2) =$$

$$(6) \quad (y - 4)(y - 6) =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (a + 8)(a - 8) =$$

$$(2) \quad (a + 6)(a - 6) =$$

$$(3) \quad (a + 7)^2 =$$

$$(4) \quad (a - 4)^2 =$$

$$(5) \quad (y + 4)^2 =$$

$$(6) \quad (x + 4)^2 =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (x - 3)^2 =$$

$$(2) \quad (a + 8)^2 =$$

$$(3) \quad (x + 7)(x - 7) =$$

$$(4) \quad (x - 5)^2 =$$

$$(5) \quad (a - 7)^2 =$$

$$(6) \quad (y + 6)^2 =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (a - 8)^2 =$$

$$(2) \quad (x - 6)^2 =$$

$$(3) \quad (y + 8)^2 =$$

$$(4) \quad (x - 4)^2 =$$

$$(5) \quad (a + 5)(a - 5) =$$

$$(6) \quad (y - 8)^2 =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (x - 3)^2 =$$

$$(2) \quad (x + 2)^2 =$$

$$(3) \quad (x - 6)^2 =$$

$$(4) \quad (a + 6)(a - 6) =$$

$$(5) \quad (a + 4)^2 =$$

$$(6) \quad (x + 6)(x - 6) =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (a - 6)^2 =$$

$$(2) \quad (x - 8)^2 =$$

$$(3) \quad (a + 3)^2 =$$

$$(4) \quad (x - 7)^2 =$$

$$(5) \quad (y - 6)^2 =$$

$$(6) \quad (a + 8)^2 =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (y + 3)(y - 3) =$$

$$(2) \quad (x - 4)^2 =$$

$$(3) \quad (x + 4)^2 =$$

$$(4) \quad (a + 7)^2 =$$

$$(5) \quad (x - 3)^2 =$$

$$(6) \quad (y + 8)^2 =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (a - 8)^2 =$$

$$(2) \quad (a + 5)^2 =$$

$$(3) \quad (a - 2)^2 =$$

$$(4) \quad (x + 4)^2 =$$

$$(5) \quad (x + 6)^2 =$$

$$(6) \quad (a + 6)(a - 6) =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (x + 4)(x - 4) =$$

$$(2) \quad (x + 4)^2 =$$

$$(3) \quad (a - 3)^2 =$$

$$(4) \quad (x - 3)^2 =$$

$$(5) \quad (y - 2)^2 =$$

$$(6) \quad (x - 5)^2 =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (x - 8)^2 =$$

$$(2) \quad (x - 4)^2 =$$

$$(3) \quad (x + 3)^2 =$$

$$(4) \quad (a - 8)^2 =$$

$$(5) \quad (y + 2)(y - 2) =$$

$$(6) \quad (y + 6)^2 =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (x + 6)(x - 6) =$$

$$(2) \quad (x - 6)^2 =$$

$$(3) \quad (a + 8)(a - 8) =$$

$$(4) \quad (a + 2)(a - 2) =$$

$$(5) \quad (y - 2)^2 =$$

$$(6) \quad (y + 4)(y - 4) =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (a - 2)^2 =$$

$$(2) \quad (x - 4)^2 =$$

$$(3) \quad (a + 7)^2 =$$

$$(4) \quad (x + 8)(x - 8) =$$

$$(5) \quad (a + 8)(a - 8) =$$

$$(6) \quad (x - 2)^2 =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (x + 7)(x - 7) =$$

$$(2) \quad (a + 3)(a - 3) =$$

$$(3) \quad (x + 4)(x - 4) =$$

$$(4) \quad (x - 6)^2 =$$

$$(5) \quad (x - 6)^2 =$$

$$(6) \quad (x - 7)^2 =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (a - 4)^2 =$$

$$(2) \quad (x + 8)(x - 8) =$$

$$(3) \quad (y + 7)(y - 7) =$$

$$(4) \quad (a - 3)^2 =$$

$$(5) \quad (x + 7)^2 =$$

$$(6) \quad (x - 4)^2 =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (a + 4)^2 =$$

$$(2) \quad (x - 8)^2 =$$

$$(3) \quad (y - 5)^2 =$$

$$(4) \quad (y + 5)(y - 5) =$$

$$(5) \quad (x - 5)^2 =$$

$$(6) \quad (y + 4)(y - 4) =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (y + 8)(y - 8) =$$

$$(2) \quad (a + 5)^2 =$$

$$(3) \quad (y + 6)^2 =$$

$$(4) \quad (a + 4)^2 =$$

$$(5) \quad (x - 2)^2 =$$

$$(6) \quad (a - 4)^2 =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (x + 8)^2 =$$

$$(2) \quad (x + 4)^2 =$$

$$(3) \quad (a - 5)^2 =$$

$$(4) \quad (y + 4)(y - 4) =$$

$$(5) \quad (a + 3)^2 =$$

$$(6) \quad (y - 8)^2 =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (y - 6)^2 =$$

$$(2) \quad (y + 3)(y - 3) =$$

$$(3) \quad (y + 4)(y - 4) =$$

$$(4) \quad (x - 7)^2 =$$

$$(5) \quad (a - 5)^2 =$$

$$(6) \quad (a - 8)^2 =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (x + 8)^2 =$$

$$(2) \quad (y + 4)(y - 4) =$$

$$(3) \quad (a - 2)^2 =$$

$$(4) \quad (x + 7)(x - 7) =$$

$$(5) \quad (y - 4)^2 =$$

$$(6) \quad (x - 6)^2 =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (a + 4)^2 =$$

$$(2) \quad (y - 7)^2 =$$

$$(3) \quad (x - 2)^2 =$$

$$(4) \quad (a + 3)^2 =$$

$$(5) \quad (a - 5)^2 =$$

$$(6) \quad (x + 4)(x - 4) =$$

Section 3. Multiplication Using Formulas

Use algebraic formulas to expand and simplify.

$$(1) \quad (x + 7)^2 =$$

$$(2) \quad (y + 4)(y - 4) =$$

$$(3) \quad (a + 7)(a - 7) =$$

$$(4) \quad (x + 3)^2 =$$

$$(5) \quad (a + 8)(a - 8) =$$

$$(6) \quad (x - 3)^2 =$$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 - 25 =$

(2) Factor: $y^2 + 6y + 8 =$

(3) Factor: $24x^2 + 6x =$

(4) Factor: $8y^2 + 16y =$

(5) Factor: $x^2 - 36 =$

(6) Factor: $x^2 - 16 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $12x^2 + 6x =$

(2) Factor: $y^2 - 16 =$

(3) Factor: $x^2 + 6x + 8 =$

(4) Factor: $x^2 + 6x + 5 =$

(5) Factor: $9y^2 + 9y =$

(6) Factor: $x^2 + 8x + 15 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 36 =$

(2) Factor: $12y^2 + 16y =$

(3) Factor: $x^2 + 7x + 10 =$

(4) Factor: $x^2 + 8x + 12 =$

(5) Factor: $y^2 + 2y + 1 =$

(6) Factor: $21x^2 + 18x =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 9 =$

(2) Factor: $6x^2 + 10x =$

(3) Factor: $y^2 - 81 =$

(4) Factor: $x^2 + 2x + 1 =$

(5) Factor: $y^2 - 81 =$

(6) Factor: $y^2 + 9y + 20 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 + 9x + 20 =$

(2) Factor: $x^2 + 10x + 24 =$

(3) Factor: $y^2 + 7y + 12 =$

(4) Factor: $8y^2 + 10y =$

(5) Factor: $x^2 + 6x + 8 =$

(6) Factor: $y^2 - 81 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 16 =$

(2) Factor: $y^2 - 25 =$

(3) Factor: $18y^2 + 15y =$

(4) Factor: $y^2 - 4 =$

(5) Factor: $12y^2 + 10y =$

(6) Factor: $6y^2 + 6y =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 - 36 =$

(2) Factor: $x^2 + 5x + 4 =$

(3) Factor: $21y^2 + 12y =$

(4) Factor: $x^2 - 64 =$

(5) Factor: $12x^2 + 18x =$

(6) Factor: $24y^2 + 20y =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $18y^2 + 9y =$

(2) Factor: $y^2 + 7y + 12 =$

(3) Factor: $12y^2 + 12y =$

(4) Factor: $x^2 - 81 =$

(5) Factor: $10y^2 + 6y =$

(6) Factor: $16x^2 + 20x =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 16 =$

(2) Factor: $16y^2 + 16y =$

(3) Factor: $x^2 - 25 =$

(4) Factor: $y^2 + 2y + 1 =$

(5) Factor: $21x^2 + 12x =$

(6) Factor: $y^2 + 7y + 10 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 - 25 =$

(2) Factor: $x^2 + 6x + 8 =$

(3) Factor: $y^2 + 8y + 12 =$

(4) Factor: $12x^2 + 4x =$

(5) Factor: $8y^2 + 10y =$

(6) Factor: $x^2 - 64 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 + 7y + 6 =$

(2) Factor: $24x^2 + 16x =$

(3) Factor: $y^2 - 49 =$

(4) Factor: $24y^2 + 16y =$

(5) Factor: $y^2 - 81 =$

(6) Factor: $12x^2 + 18x =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 + 8x + 15 =$

(2) Factor: $x^2 - 64 =$

(3) Factor: $y^2 + 8y + 15 =$

(4) Factor: $y^2 + 7y + 12 =$

(5) Factor: $12y^2 + 6y =$

(6) Factor: $y^2 + 6y + 5 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $32y^2 + 16y =$

(2) Factor: $12y^2 + 15y =$

(3) Factor: $y^2 + 4y + 3 =$

(4) Factor: $y^2 + 7y + 12 =$

(5) Factor: $8y^2 + 16y =$

(6) Factor: $20y^2 + 16y =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $21y^2 + 15y =$

(2) Factor: $x^2 + 6x + 9 =$

(3) Factor: $x^2 - 9 =$

(4) Factor: $x^2 + 3x + 2 =$

(5) Factor: $x^2 - 4 =$

(6) Factor: $y^2 - 16 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 + 2y + 1 =$

(2) Factor: $x^2 - 49 =$

(3) Factor: $y^2 - 81 =$

(4) Factor: $8y^2 + 6y =$

(5) Factor: $x^2 - 81 =$

(6) Factor: $y^2 + 3y + 2 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 25 =$

(2) Factor: $y^2 + 5y + 6 =$

(3) Factor: $12x^2 + 8x =$

(4) Factor: $x^2 + 3x + 2 =$

(5) Factor: $8y^2 + 10y =$

(6) Factor: $x^2 + 6x + 5 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 + 8y + 15 =$

(2) Factor: $x^2 + 7x + 10 =$

(3) Factor: $y^2 + 9y + 20 =$

(4) Factor: $x^2 - 64 =$

(5) Factor: $x^2 + 4x + 4 =$

(6) Factor: $y^2 - 25 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 36 =$

(2) Factor: $x^2 - 25 =$

(3) Factor: $12y^2 + 20y =$

(4) Factor: $y^2 + 10y + 24 =$

(5) Factor: $y^2 + 4y + 4 =$

(6) Factor: $x^2 + 9x + 20 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 + 6y + 5 =$

(2) Factor: $32x^2 + 24x =$

(3) Factor: $12x^2 + 16x =$

(4) Factor: $x^2 - 25 =$

(5) Factor: $y^2 + 6y + 5 =$

(6) Factor: $x^2 - 81 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $15y^2 + 9y =$

(2) Factor: $y^2 - 36 =$

(3) Factor: $x^2 + 9x + 18 =$

(4) Factor: $x^2 - 4 =$

(5) Factor: $x^2 + 7x + 12 =$

(6) Factor: $28y^2 + 24y =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $10y^2 + 12y =$

(2) Factor: $y^2 + 9y + 18 =$

(3) Factor: $16y^2 + 8y =$

(4) Factor: $6x^2 + 15x =$

(5) Factor: $x^2 + 2x + 1 =$

(6) Factor: $x^2 - 36 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 - 64 =$

(2) Factor: $10x^2 + 8x =$

(3) Factor: $x^2 - 9 =$

(4) Factor: $12y^2 + 4y =$

(5) Factor: $4x^2 + 12x =$

(6) Factor: $x^2 - 25 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 + 8x + 16 =$

(2) Factor: $16x^2 + 4x =$

(3) Factor: $y^2 + 8y + 12 =$

(4) Factor: $y^2 + 5y + 6 =$

(5) Factor: $x^2 - 25 =$

(6) Factor: $y^2 - 36 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 - 16 =$

(2) Factor: $y^2 + 7y + 12 =$

(3) Factor: $y^2 - 64 =$

(4) Factor: $14x^2 + 8x =$

(5) Factor: $24y^2 + 12y =$

(6) Factor: $24x^2 + 6x =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $28x^2 + 20x =$

(2) Factor: $x^2 - 49 =$

(3) Factor: $y^2 + 9y + 18 =$

(4) Factor: $y^2 - 16 =$

(5) Factor: $y^2 - 16 =$

(6) Factor: $y^2 + 4y + 4 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $12x^2 + 24x =$

(2) Factor: $x^2 + 6x + 8 =$

(3) Factor: $y^2 - 4 =$

(4) Factor: $28x^2 + 8x =$

(5) Factor: $x^2 - 25 =$

(6) Factor: $x^2 - 49 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 - 81 =$

(2) Factor: $y^2 + 6y + 9 =$

(3) Factor: $y^2 - 81 =$

(4) Factor: $16y^2 + 8y =$

(5) Factor: $y^2 - 64 =$

(6) Factor: $14x^2 + 6x =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 - 81 =$

(2) Factor: $20y^2 + 24y =$

(3) Factor: $6y^2 + 8y =$

(4) Factor: $x^2 - 81 =$

(5) Factor: $x^2 - 25 =$

(6) Factor: $20y^2 + 24y =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 - 36 =$

(2) Factor: $12x^2 + 9x =$

(3) Factor: $x^2 + 9x + 20 =$

(4) Factor: $y^2 + 7y + 12 =$

(5) Factor: $y^2 + 6y + 8 =$

(6) Factor: $y^2 + 4y + 4 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $8y^2 + 8y =$

(2) Factor: $y^2 + 4y + 3 =$

(3) Factor: $y^2 + 3y + 2 =$

(4) Factor: $12x^2 + 12x =$

(5) Factor: $y^2 + 6y + 9 =$

(6) Factor: $x^2 - 49 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 - 25 =$

(2) Factor: $x^2 + 9x + 18 =$

(3) Factor: $6x^2 + 15x =$

(4) Factor: $y^2 - 4 =$

(5) Factor: $x^2 + 3x + 2 =$

(6) Factor: $y^2 - 49 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 + 7y + 10 =$

(2) Factor: $x^2 + 2x + 1 =$

(3) Factor: $x^2 + 5x + 4 =$

(4) Factor: $4x^2 + 6x =$

(5) Factor: $16y^2 + 8y =$

(6) Factor: $y^2 + 5y + 4 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 + 3x + 2 =$

(2) Factor: $8y^2 + 4y =$

(3) Factor: $y^2 + 3y + 2 =$

(4) Factor: $x^2 - 25 =$

(5) Factor: $x^2 + 5x + 4 =$

(6) Factor: $x^2 - 25 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $18y^2 + 15y =$

(2) Factor: $6y^2 + 18y =$

(3) Factor: $4x^2 + 8x =$

(4) Factor: $x^2 - 64 =$

(5) Factor: $y^2 - 9 =$

(6) Factor: $12x^2 + 15x =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 - 16 =$

(2) Factor: $20x^2 + 20x =$

(3) Factor: $y^2 + 2y + 1 =$

(4) Factor: $24y^2 + 8y =$

(5) Factor: $y^2 + 5y + 4 =$

(6) Factor: $y^2 - 64 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 + 7y + 12 =$

(2) Factor: $x^2 + 6x + 5 =$

(3) Factor: $y^2 + 2y + 1 =$

(4) Factor: $x^2 - 9 =$

(5) Factor: $y^2 - 49 =$

(6) Factor: $24y^2 + 15y =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $12x^2 + 12x =$

(2) Factor: $y^2 + 4y + 4 =$

(3) Factor: $y^2 - 36 =$

(4) Factor: $x^2 - 9 =$

(5) Factor: $12y^2 + 15y =$

(6) Factor: $x^2 - 49 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $y^2 - 16 =$

(2) Factor: $y^2 - 36 =$

(3) Factor: $y^2 + 5y + 6 =$

(4) Factor: $y^2 - 64 =$

(5) Factor: $x^2 - 25 =$

(6) Factor: $x^2 + 7x + 6 =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 - 81 =$

(2) Factor: $x^2 - 64 =$

(3) Factor: $y^2 + 8y + 12 =$

(4) Factor: $x^2 - 36 =$

(5) Factor: $x^2 + 8x + 16 =$

(6) Factor: $24y^2 + 20y =$

Section 5. Factorisation II

Factor each expression completely.

(1) Factor: $x^2 - 81 =$

(2) Factor: $x^2 + 5x + 4 =$

(3) Factor: $6y^2 + 15y =$

(4) Factor: $y^2 + 7y + 12 =$

(5) Factor: $y^2 - 16 =$

(6) Factor: $y^2 - 25 =$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{45} =$

(2) $2\sqrt{7} + 2\sqrt{7} =$

(3) $2\sqrt{3} + \sqrt{3} =$

(4) $4\sqrt{7} + \sqrt{7} =$

(5) $\sqrt{64} =$

(6) $\sqrt{45} =$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{12} =$

(2) $\sqrt{45} =$

(3) $\sqrt{9} =$

(4) $3\sqrt{5} + 2\sqrt{5} =$

(5) $\sqrt{50} =$

(6) $\sqrt{100} =$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{32} =$

(2) $\sqrt{12} =$

(3) $\sqrt{50} =$

(4) $\sqrt{12} =$

(5) $\sqrt{8} =$

(6) $2\sqrt{5} + \sqrt{5} =$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{4} =$

(2) $\sqrt{36} =$

(3) $\sqrt{100} =$

(4) $3\sqrt{5} + 2\sqrt{5} =$

(5) $\sqrt{50} =$

(6) $3\sqrt{2} + 2\sqrt{2} =$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad 2\sqrt{3} + 3\sqrt{3} =$$

$$(2) \quad 3\sqrt{3} + 3\sqrt{3} =$$

$$(3) \quad 3\sqrt{7} + 2\sqrt{7} =$$

$$(4) \quad \sqrt{50} =$$

$$(5) \quad \sqrt{64} =$$

$$(6) \quad 3\sqrt{6} + \sqrt{6} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad 3\sqrt{5} + 2\sqrt{5} =$$

$$(2) \quad \sqrt{81} =$$

$$(3) \quad \sqrt{36} =$$

$$(4) \quad 5\sqrt{5} + 3\sqrt{5} =$$

$$(5) \quad 5\sqrt{6} + 3\sqrt{6} =$$

$$(6) \quad 2\sqrt{5} + 2\sqrt{5} =$$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{18} =$

(2) $4\sqrt{7} + 3\sqrt{7} =$

(3) $\sqrt{16} =$

(4) $\sqrt{12} =$

(5) $\sqrt{36} =$

(6) $\sqrt{16} =$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{8} =$

(2) $3\sqrt{7} + 2\sqrt{7} =$

(3) $\sqrt{45} =$

(4) $\sqrt{25} =$

(5) $\sqrt{36} =$

(6) $2\sqrt{2} + 3\sqrt{2} =$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{18} =$

(2) $5\sqrt{7} + 2\sqrt{7} =$

(3) $\sqrt{64} =$

(4) $\sqrt{18} =$

(5) $\sqrt{18} =$

(6) $3\sqrt{5} + 2\sqrt{5} =$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{50} =$

(2) $\sqrt{18} =$

(3) $\sqrt{4} =$

(4) $\sqrt{36} =$

(5) $\sqrt{36} =$

(6) $\sqrt{100} =$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{18} =$

(2) $\sqrt{50} =$

(3) $2\sqrt{2} + 3\sqrt{2} =$

(4) $\sqrt{32} =$

(5) $2\sqrt{6} + 3\sqrt{6} =$

(6) $3\sqrt{2} + 2\sqrt{2} =$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{16} =$

(2) $\sqrt{50} =$

(3) $2\sqrt{2} + 2\sqrt{2} =$

(4) $\sqrt{18} =$

(5) $\sqrt{64} =$

(6) $5\sqrt{6} + 2\sqrt{6} =$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad 3\sqrt{2} + 2\sqrt{2} =$$

$$(2) \quad \sqrt{27} =$$

$$(3) \quad \sqrt{12} =$$

$$(4) \quad \sqrt{12} =$$

$$(5) \quad \sqrt{25} =$$

$$(6) \quad 4\sqrt{7} + 3\sqrt{7} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad 2\sqrt{6} + \sqrt{6} =$$

$$(2) \quad \sqrt{36} =$$

$$(3) \quad 3\sqrt{3} + 3\sqrt{3} =$$

$$(4) \quad \sqrt{16} =$$

$$(5) \quad 4\sqrt{6} + \sqrt{6} =$$

$$(6) \quad \sqrt{18} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad 4\sqrt{6} + 3\sqrt{6} =$$

$$(2) \quad \sqrt{50} =$$

$$(3) \quad \sqrt{9} =$$

$$(4) \quad \sqrt{27} =$$

$$(5) \quad 4\sqrt{6} + 2\sqrt{6} =$$

$$(6) \quad \sqrt{8} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad 5\sqrt{7} + \sqrt{7} =$$

$$(2) \quad \sqrt{25} =$$

$$(3) \quad \sqrt{4} =$$

$$(4) \quad 4\sqrt{5} + 3\sqrt{5} =$$

$$(5) \quad 3\sqrt{5} + 2\sqrt{5} =$$

$$(6) \quad \sqrt{27} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad 2\sqrt{6} + 3\sqrt{6} =$$

$$(2) \quad 4\sqrt{7} + \sqrt{7} =$$

$$(3) \quad \sqrt{20} =$$

$$(4) \quad \sqrt{20} =$$

$$(5) \quad \sqrt{27} =$$

$$(6) \quad \sqrt{27} =$$

Section 6. Square Roots

Simplify the square root expressions.

$$(1) \quad \sqrt{20} =$$

$$(2) \quad 3\sqrt{5} + 3\sqrt{5} =$$

$$(3) \quad \sqrt{36} =$$

$$(4) \quad \sqrt{49} =$$

$$(5) \quad \sqrt{12} =$$

$$(6) \quad \sqrt{18} =$$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{64} =$

(2) $4\sqrt{3} + 3\sqrt{3} =$

(3) $\sqrt{12} =$

(4) $2\sqrt{3} + \sqrt{3} =$

(5) $\sqrt{12} =$

(6) $\sqrt{25} =$

Section 6. Square Roots

Simplify the square root expressions.

(1) $\sqrt{18} =$

(2) $5\sqrt{2} + 2\sqrt{2} =$

(3) $\sqrt{9} =$

(4) $5\sqrt{2} + \sqrt{2} =$

(5) $\sqrt{81} =$

(6) $\sqrt{8} =$

Section 7. Quadratic Equations I

Solve each quadratic equation.

(1) $2y^2 - 2 = 0$

(2) $2y^2 - 14 = 0$

(3) $3x^2 - 6 = 0$

(4) $y^2 + 2y + 1 = 0$

(5) $y^2 + 6y + 9 = 0$

(6) $y^2 + 4y + 4 = 0$

Section 7. Quadratic Equations I

Solve each quadratic equation.

(1) $2x^2 - 10 = 0$

(2) $4x^2 - 20 = 0$

(3) $x^2 + 7x + 12 = 0$

(4) $2x^2 - 16 = 0$

(5) $x^2 + 6x = 8$

(6) $4x^2 - 4 = 0$

Section 7. Quadratic Equations I

Solve each quadratic equation.

$$(1) \quad y^2 + 6y = 7$$

$$(2) \quad 3y^2 - 9 = 0$$

$$(3) \quad x^2 + 4x = 10$$

$$(4) \quad x^2 + 5x + 6 = 0$$

$$(5) \quad x^2 + 7x + 12 = 0$$

$$(6) \quad 3y^2 - 21 = 0$$

Section 7. Quadratic Equations I

Solve each quadratic equation.

$$(1) \quad x^2 + 6x + 8 = 0$$

$$(2) \quad x^2 + 14x = 3$$

$$(3) \quad y^2 + 16y = 4$$

$$(4) \quad x^2 + 3x + 2 = 0$$

$$(5) \quad 3y^2 - 21 = 0$$

$$(6) \quad x^2 + 6x = 5$$

Section 7. Quadratic Equations I

Solve each quadratic equation.

$$(1) \quad x^2 + 4x + 3 = 0$$

$$(2) \quad x^2 + 12x = 4$$

$$(3) \quad y^2 + 5y + 4 = 0$$

$$(4) \quad y^2 + 14y = 4$$

$$(5) \quad x^2 + 5x + 4 = 0$$

$$(6) \quad y^2 + 4y + 3 = 0$$

Section 7. Quadratic Equations I

Solve each quadratic equation.

$$(1) \quad x^2 + 10x = 8$$

$$(2) \quad 2y^2 - 16 = 0$$

$$(3) \quad 3x^2 - 9 = 0$$

$$(4) \quad x^2 + 12x = 2$$

$$(5) \quad x^2 + 2x + 1 = 0$$

$$(6) \quad x^2 + 7x + 12 = 0$$

Section 7. Quadratic Equations I

Solve each quadratic equation.

$$(1) \quad x^2 + 6x = 1$$

$$(2) \quad 2y^2 - 12 = 0$$

$$(3) \quad y^2 + 6y + 9 = 0$$

$$(4) \quad y^2 + 2y + 1 = 0$$

$$(5) \quad x^2 + 5x + 4 = 0$$

$$(6) \quad y^2 + 6y + 9 = 0$$

Section 7. Quadratic Equations I

Solve each quadratic equation.

$$(1) \quad 2x^2 - 2 = 0$$

$$(2) \quad 4x^2 - 4 = 0$$

$$(3) \quad y^2 + 6y + 9 = 0$$

$$(4) \quad y^2 + 5y + 6 = 0$$

$$(5) \quad 2x^2 - 12 = 0$$

$$(6) \quad y^2 + 12y = 2$$

Section 7. Quadratic Equations I

Solve each quadratic equation.

(1) $x^2 + 12x = 8$

(2) $y^2 + 5y + 6 = 0$

(3) $4y^2 - 16 = 0$

(4) $y^2 + 12y = 8$

(5) $x^2 + 3x + 2 = 0$

(6) $x^2 + 6x = 9$

Section 7. Quadratic Equations I

Solve each quadratic equation.

(1) $4x^2 - 12 = 0$

(2) $2y^2 - 2 = 0$

(3) $x^2 + 6x = 5$

(4) $y^2 + 16y = 7$

(5) $3y^2 - 24 = 0$

(6) $x^2 + 4x = 4$

Section 7. Quadratic Equations I

Solve each quadratic equation.

$$(1) \quad x^2 + 3x + 2 = 0$$

$$(2) \quad y^2 + 7y + 12 = 0$$

$$(3) \quad x^2 + 12x = 6$$

$$(4) \quad y^2 + 7y + 12 = 0$$

$$(5) \quad x^2 + 4x = 3$$

$$(6) \quad 4y^2 - 4 = 0$$

Section 7. Quadratic Equations I

Solve each quadratic equation.

$$(1) \quad x^2 + 8x = 1$$

$$(2) \quad 4y^2 - 20 = 0$$

$$(3) \quad x^2 + 6x + 8 = 0$$

$$(4) \quad x^2 + 6x = 5$$

$$(5) \quad x^2 + 6x + 8 = 0$$

$$(6) \quad 3y^2 - 6 = 0$$

Section 7. Quadratic Equations I

Solve each quadratic equation.

$$(1) \quad y^2 + 3y + 2 = 0$$

$$(2) \quad 3x^2 - 21 = 0$$

$$(3) \quad 4x^2 - 20 = 0$$

$$(4) \quad x^2 + 4x = 7$$

$$(5) \quad x^2 + 4x = 8$$

$$(6) \quad x^2 + 2x + 1 = 0$$

Section 7. Quadratic Equations I

Solve each quadratic equation.

$$(1) \quad y^2 + 10y = 1$$

$$(2) \quad y^2 + 6y = 6$$

$$(3) \quad x^2 + 6x + 9 = 0$$

$$(4) \quad x^2 + 4x = 10$$

$$(5) \quad 4x^2 - 20 = 0$$

$$(6) \quad y^2 + 12y = 8$$

Section 7. Quadratic Equations I

Solve each quadratic equation.

$$(1) \quad x^2 + 7x + 12 = 0$$

$$(2) \quad y^2 + 4y + 4 = 0$$

$$(3) \quad 3x^2 - 18 = 0$$

$$(4) \quad y^2 + 8y = 10$$

$$(5) \quad 3y^2 - 15 = 0$$

$$(6) \quad x^2 + 5x + 6 = 0$$

Section 7. Quadratic Equations I

Solve each quadratic equation.

$$(1) \quad x^2 + 4x + 4 = 0$$

$$(2) \quad y^2 + 14y = 2$$

$$(3) \quad 2y^2 - 2 = 0$$

$$(4) \quad x^2 + 6x + 8 = 0$$

$$(5) \quad y^2 + 2y + 1 = 0$$

$$(6) \quad x^2 + 6x = 10$$

Section 7. Quadratic Equations I

Solve each quadratic equation.

$$(1) \quad y^2 + 5y + 4 = 0$$

$$(2) \quad 4y^2 - 16 = 0$$

$$(3) \quad x^2 + 4x + 4 = 0$$

$$(4) \quad x^2 + 16x = 4$$

$$(5) \quad 3y^2 - 9 = 0$$

$$(6) \quad x^2 + 6x + 9 = 0$$

Section 7. Quadratic Equations I

Solve each quadratic equation.

$$(1) \quad x^2 + 16x = 6$$

$$(2) \quad x^2 + 4x + 4 = 0$$

$$(3) \quad y^2 + 4y + 3 = 0$$

$$(4) \quad 4y^2 - 8 = 0$$

$$(5) \quad y^2 + 8y = 3$$

$$(6) \quad y^2 + 7y + 12 = 0$$

Section 7. Quadratic Equations I

Solve each quadratic equation.

(1) $y^2 + 4y + 4 = 0$

(2) $y^2 + 14y = 5$

(3) $x^2 + 5x + 4 = 0$

(4) $2x^2 - 12 = 0$

(5) $4x^2 - 8 = 0$

(6) $x^2 + 8x = 2$

Section 7. Quadratic Equations I

Solve each quadratic equation.

(1) $4x^2 - 24 = 0$

(2) $4x^2 - 16 = 0$

(3) $y^2 + 6y = 8$

(4) $x^2 + 3x + 2 = 0$

(5) $2x^2 - 16 = 0$

(6) $y^2 + 4y + 4 = 0$

Section 8. Quadratic Equations II

Solve each quadratic equation.

$$(1) \quad y^2 + 5y + 6 = 0$$

$$(2) \quad 4x^2 - 24 = 0$$

$$(3) \quad x^2 + 8x = 4$$

$$(4) \quad x^2 + 6x + 9 = 0$$

$$(5) \quad x^2 + 5x + 4 = 0$$

$$(6) \quad y^2 + 6y = 7$$

Section 8. Quadratic Equations II

Solve each quadratic equation.

$$(1) \quad 4x^2 - 8 = 0$$

$$(2) \quad y^2 + 14y = 1$$

$$(3) \quad x^2 + 12x = 7$$

$$(4) \quad x^2 + 4x + 3 = 0$$

$$(5) \quad x^2 + 7x + 12 = 0$$

$$(6) \quad 3y^2 - 12 = 0$$

Section 8. Quadratic Equations II

Solve each quadratic equation.

$$(1) \quad x^2 + 8x + 16 = 0$$

$$(2) \quad x^2 + 4x + 3 = 0$$

$$(3) \quad x^2 + 8x = 10$$

$$(4) \quad x^2 + 4x = 1$$

$$(5) \quad x^2 + 4x + 3 = 0$$

$$(6) \quad x^2 + 8x = 7$$

Section 8. Quadratic Equations II

Solve each quadratic equation.

$$(1) \quad y^2 + 8y = 9$$

$$(2) \quad x^2 + 16x = 8$$

$$(3) \quad 2x^2 - 16 = 0$$

$$(4) \quad x^2 + 4x + 3 = 0$$

$$(5) \quad y^2 + 4y = 2$$

$$(6) \quad x^2 + 10x = 3$$

Section 8. Quadratic Equations II

Solve each quadratic equation.

$$(1) \quad y^2 + 14y = 5$$

$$(2) \quad y^2 + 5y + 6 = 0$$

$$(3) \quad y^2 + 6y = 8$$

$$(4) \quad y^2 + 8y + 16 = 0$$

$$(5) \quad x^2 + 16x = 5$$

$$(6) \quad 2x^2 - 4 = 0$$

Section 8. Quadratic Equations II

Solve each quadratic equation.

$$(1) \quad x^2 + 3x + 2 = 0$$

$$(2) \quad x^2 + 14x = 7$$

$$(3) \quad x^2 + 6x + 9 = 0$$

$$(4) \quad 2x^2 - 14 = 0$$

$$(5) \quad 4y^2 - 24 = 0$$

$$(6) \quad y^2 + 16y = 1$$

Section 8. Quadratic Equations II

Solve each quadratic equation.

(1) $2y^2 - 14 = 0$

(2) $x^2 + 2x + 1 = 0$

(3) $3x^2 - 18 = 0$

(4) $y^2 + 10y = 8$

(5) $2y^2 - 6 = 0$

(6) $y^2 + 6y = 10$

Section 8. Quadratic Equations II

Solve each quadratic equation.

(1) $y^2 + 6y + 8 = 0$

(2) $3x^2 - 21 = 0$

(3) $4y^2 - 16 = 0$

(4) $y^2 + 4y = 7$

(5) $4x^2 - 12 = 0$

(6) $y^2 + 4y + 4 = 0$

Section 8. Quadratic Equations II

Solve each quadratic equation.

$$(1) \quad 2y^2 - 14 = 0$$

$$(2) \quad x^2 + 4x = 8$$

$$(3) \quad y^2 + 4y = 6$$

$$(4) \quad y^2 + 8y = 8$$

$$(5) \quad 2x^2 - 6 = 0$$

$$(6) \quad y^2 + 16y = 6$$

Section 8. Quadratic Equations II

Solve each quadratic equation.

$$(1) \quad x^2 + 5x + 4 = 0$$

$$(2) \quad x^2 + 8x = 6$$

$$(3) \quad y^2 + 14y = 2$$

$$(4) \quad 2x^2 - 12 = 0$$

$$(5) \quad y^2 + 6y = 8$$

$$(6) \quad 3y^2 - 9 = 0$$

Section 8. Quadratic Equations II

Solve each quadratic equation.

(1) $x^2 + 8x = 1$

(2) $x^2 + 6x = 10$

(3) $x^2 + 4x + 4 = 0$

(4) $y^2 + 12y = 1$

(5) $3x^2 - 9 = 0$

(6) $4x^2 - 4 = 0$

Section 8. Quadratic Equations II

Solve each quadratic equation.

(1) $y^2 + 12y = 2$

(2) $y^2 + 6y + 8 = 0$

(3) $4y^2 - 24 = 0$

(4) $x^2 + 4x + 3 = 0$

(5) $2y^2 - 2 = 0$

(6) $x^2 + 3x + 2 = 0$

Section 8. Quadratic Equations II

Solve each quadratic equation.

$$(1) \quad y^2 + 6y + 9 = 0$$

$$(2) \quad y^2 + 16y = 6$$

$$(3) \quad y^2 + 6y = 8$$

$$(4) \quad y^2 + 12y = 2$$

$$(5) \quad y^2 + 6y + 9 = 0$$

$$(6) \quad 3x^2 - 6 = 0$$

Section 8. Quadratic Equations II

Solve each quadratic equation.

$$(1) \quad y^2 + 10y = 1$$

$$(2) \quad 3x^2 - 12 = 0$$

$$(3) \quad 4y^2 - 24 = 0$$

$$(4) \quad 2y^2 - 6 = 0$$

$$(5) \quad x^2 + 10x = 3$$

$$(6) \quad 4y^2 - 12 = 0$$

Section 8. Quadratic Equations II

Solve each quadratic equation.

$$(1) \quad 4y^2 - 8 = 0$$

$$(2) \quad y^2 + 5y + 6 = 0$$

$$(3) \quad y^2 + 8y = 5$$

$$(4) \quad 3y^2 - 6 = 0$$

$$(5) \quad x^2 + 16x = 10$$

$$(6) \quad 3x^2 - 12 = 0$$

Section 8. Quadratic Equations II

Solve each quadratic equation.

$$(1) \quad y^2 + 14y = 6$$

$$(2) \quad y^2 + 14y = 5$$

$$(3) \quad x^2 + 3x + 2 = 0$$

$$(4) \quad 4y^2 - 24 = 0$$

$$(5) \quad 4x^2 - 4 = 0$$

$$(6) \quad 3y^2 - 15 = 0$$

Section 8. Quadratic Equations II

Solve each quadratic equation.

$$(1) \quad 4x^2 - 28 = 0$$

$$(2) \quad 3y^2 - 3 = 0$$

$$(3) \quad x^2 + 5x + 6 = 0$$

$$(4) \quad y^2 + 6y = 6$$

$$(5) \quad 2x^2 - 16 = 0$$

$$(6) \quad x^2 + 8x = 1$$

Section 8. Quadratic Equations II

Solve each quadratic equation.

$$(1) \quad 2x^2 - 12 = 0$$

$$(2) \quad x^2 + 4x = 7$$

$$(3) \quad 2y^2 - 4 = 0$$

$$(4) \quad y^2 + 7y + 12 = 0$$

$$(5) \quad 3y^2 - 9 = 0$$

$$(6) \quad y^2 + 8y = 4$$

Section 8. Quadratic Equations II

Solve each quadratic equation.

$$(1) \quad y^2 + 3y + 2 = 0$$

$$(2) \quad x^2 + 3x + 2 = 0$$

$$(3) \quad 2x^2 - 12 = 0$$

$$(4) \quad 3x^2 - 15 = 0$$

$$(5) \quad 4x^2 - 4 = 0$$

$$(6) \quad 4x^2 - 4 = 0$$

Section 8. Quadratic Equations II

Solve each quadratic equation.

$$(1) \quad y^2 + 5y + 6 = 0$$

$$(2) \quad x^2 + 5x + 4 = 0$$

$$(3) \quad y^2 + 8y + 16 = 0$$

$$(4) \quad y^2 + 10y = 1$$

$$(5) \quad x^2 + 4x + 3 = 0$$

$$(6) \quad x^2 + 4x = 1$$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 14x + 2$

(2) Find x-intercepts of $y = x^2 - 25$

(3) Find x-intercepts of $y = x^2 - 4$

(4) $y = -x^2$ opens:

(5) Find x-intercepts of $y = x^2 - 9$

(6) Find vertex of $y = x^2 + 14x + 1$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 8x + 2$

(2) Find vertex of $y = x^2 + 10x + 4$

(3) $y = -2x^2$ opens:

(4) Find x-intercepts of $y = x^2 - 9$

(5) Find x-intercepts of $y = x^2 - 25$

(6) Find x-intercepts of $y = x^2 - 9$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 12x + 8$

(2) Find vertex of $y = x^2 + 14x + 8$

(3) Find x-intercepts of $y = x^2 - 9$

(4) Find vertex of $y = x^2 + 4x + 2$

(5) Find vertex of $y = x^2 + 16x + 8$

(6) Find vertex of $y = x^2 + 4x + 9$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find x-intercepts of $y = x^2 - 25$

(2) $y = -3x^2$ opens:

(3) $y = 3x^2$ opens:

(4) Find x-intercepts of $y = x^2 - 9$

(5) Find x-intercepts of $y = x^2 - 4$

(6) $y = 3x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 14x + 2$

(2) Find x-intercepts of $y = x^2 - 9$

(3) Find x-intercepts of $y = x^2 - 16$

(4) Find x-intercepts of $y = x^2 - 25$

(5) Find vertex of $y = x^2 + 4x + 3$

(6) Find vertex of $y = x^2 + 16x + 5$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find x-intercepts of $y = x^2 - 16$

(2) $y = -3x^2$ opens:

(3) Find x-intercepts of $y = x^2 - 16$

(4) Find vertex of $y = x^2 + 4x + 7$

(5) Find x-intercepts of $y = x^2 - 25$

(6) Find x-intercepts of $y = x^2 - 25$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = 2x^2$ opens:

(2) Find x-intercepts of $y = x^2 - 9$

(3) Find vertex of $y = x^2 + 6x + 1$

(4) Find vertex of $y = x^2 + 4x + 7$

(5) $y = 2x^2$ opens:

(6) $y = 2x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find x-intercepts of $y = x^2 - 16$

(2) Find x-intercepts of $y = x^2 - 4$

(3) Find vertex of $y = x^2 + 16x + 1$

(4) Find vertex of $y = x^2 + 16x + 3$

(5) $y = -2x^2$ opens:

(6) Find x-intercepts of $y = x^2 - 16$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 12x + 7$

(2) Find x-intercepts of $y = x^2 - 4$

(3) $y = -2x^2$ opens:

(4) Find x-intercepts of $y = x^2 - 25$

(5) Find vertex of $y = x^2 + 10x + 8$

(6) Find vertex of $y = x^2 + 4x + 5$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = 3x^2$ opens:

(2) Find vertex of $y = x^2 + 10x + 4$

(3) Find x-intercepts of $y = x^2 - 16$

(4) $y = 2x^2$ opens:

(5) Find vertex of $y = x^2 + 16x + 9$

(6) $y = -3x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 14x + 3$

(2) $y = 2x^2$ opens:

(3) Find x-intercepts of $y = x^2 - 9$

(4) Find vertex of $y = x^2 + 16x + 8$

(5) Find x-intercepts of $y = x^2 - 9$

(6) Find x-intercepts of $y = x^2 - 16$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = -x^2$ opens:

(2) Find vertex of $y = x^2 + 14x + 9$

(3) $y = 3x^2$ opens:

(4) $y = 3x^2$ opens:

(5) Find vertex of $y = x^2 + 14x + 6$

(6) $y = 3x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 12x + 4$

(2) $y = 2x^2$ opens:

(3) Find x-intercepts of $y = x^2 - 9$

(4) Find vertex of $y = x^2 + 10x + 8$

(5) $y = 2x^2$ opens:

(6) Find vertex of $y = x^2 + 16x + 8$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = -x^2$ opens:

(2) $y = 3x^2$ opens:

(3) Find vertex of $y = x^2 + 14x + 8$

(4) Find x-intercepts of $y = x^2 - 16$

(5) Find vertex of $y = x^2 + 16x + 4$

(6) Find x-intercepts of $y = x^2 - 16$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = 3x^2$ opens:

(2) $y = 2x^2$ opens:

(3) $y = -x^2$ opens:

(4) $y = 2x^2$ opens:

(5) Find vertex of $y = x^2 + 4x + 4$

(6) Find vertex of $y = x^2 + 8x + 6$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find x-intercepts of $y = x^2 - 16$

(2) Find vertex of $y = x^2 + 10x + 1$

(3) Find vertex of $y = x^2 + 10x + 7$

(4) Find vertex of $y = x^2 + 12x + 9$

(5) Find vertex of $y = x^2 + 16x + 2$

(6) $y = 3x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = -x^2$ opens:

(2) $y = 3x^2$ opens:

(3) Find x-intercepts of $y = x^2 - 4$

(4) Find vertex of $y = x^2 + 6x + 8$

(5) Find vertex of $y = x^2 + 16x + 7$

(6) Find vertex of $y = x^2 + 8x + 4$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = 3x^2$ opens:

(2) $y = -2x^2$ opens:

(3) Find x-intercepts of $y = x^2 - 4$

(4) $y = -3x^2$ opens:

(5) $y = 2x^2$ opens:

(6) $y = 2x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = -2x^2$ opens:

(2) $y = 3x^2$ opens:

(3) Find vertex of $y = x^2 + 14x + 3$

(4) Find vertex of $y = x^2 + 10x + 6$

(5) $y = 2x^2$ opens:

(6) Find x-intercepts of $y = x^2 - 9$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 4x + 7$

(2) Find vertex of $y = x^2 + 10x + 4$

(3) $y = -x^2$ opens:

(4) Find vertex of $y = x^2 + 16x + 7$

(5) Find vertex of $y = x^2 + 14x + 2$

(6) Find x-intercepts of $y = x^2 - 25$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = 8, b = ?, c = 17$

(2) Right triangle: $a = 8, b = ?, c = 17$

(3) Right triangle: $a = ?, b = 24, c = 25$

(4) Right triangle: $a = 5, b = ?, c = 13$

(5) Right triangle: $a = 6, b = 8, c = ?$

(6) Right triangle: $a = 9, b = 12, c = ?$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = 9, b = ?, c = 15$

(2) Right triangle: $a = 9, b = ?, c = 15$

(3) Right triangle: $a = ?, b = 12, c = 15$

(4) Right triangle: $a = 9, b = 12, c = ?$

(5) Right triangle: $a = 5, b = 12, c = ?$

(6) Right triangle: $a = 6, b = 8, c = ?$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = 9, b = ?, c = 15$

(2) Right triangle: $a = ?, b = 12, c = 15$

(3) Right triangle: $a = 8, b = 15, c = ?$

(4) Right triangle: $a = 7, b = 24, c = ?$

(5) Right triangle: $a = 6, b = 8, c = ?$

(6) Right triangle: $a = 9, b = ?, c = 15$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = 8, b = 15, c = ?$

(2) Right triangle: $a = 6, b = ?, c = 10$

(3) Right triangle: $a = 9, b = 12, c = ?$

(4) Right triangle: $a = 6, b = 8, c = ?$

(5) Right triangle: $a = ?, b = 12, c = 13$

(6) Right triangle: $a = 5, b = 12, c = ?$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = 9, b = 12, c = ?$

(2) Right triangle: $a = 6, b = ?, c = 10$

(3) Right triangle: $a = ?, b = 24, c = 25$

(4) Right triangle: $a = ?, b = 8, c = 10$

(5) Right triangle: $a = 3, b = ?, c = 5$

(6) Right triangle: $a = ?, b = 12, c = 13$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = 5, b = 12, c = ?$

(2) Right triangle: $a = 3, b = ?, c = 5$

(3) Right triangle: $a = ?, b = 24, c = 25$

(4) Right triangle: $a = 6, b = 8, c = ?$

(5) Right triangle: $a = 9, b = 12, c = ?$

(6) Right triangle: $a = 5, b = 12, c = ?$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = 9, b = 12, c = ?$

(2) Right triangle: $a = ?, b = 24, c = 25$

(3) Right triangle: $a = 6, b = 8, c = ?$

(4) Right triangle: $a = 8, b = 15, c = ?$

(5) Right triangle: $a = ?, b = 12, c = 15$

(6) Right triangle: $a = 7, b = ?, c = 25$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = ?, b = 12, c = 15$

(2) Right triangle: $a = 5, b = ?, c = 13$

(3) Right triangle: $a = 5, b = ?, c = 13$

(4) Right triangle: $a = 9, b = ?, c = 15$

(5) Right triangle: $a = ?, b = 12, c = 13$

(6) Right triangle: $a = 9, b = ?, c = 15$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = 7, b = 24, c = ?$

(2) Right triangle: $a = 8, b = 15, c = ?$

(3) Right triangle: $a = ?, b = 4, c = 5$

(4) Right triangle: $a = 9, b = ?, c = 15$

(5) Right triangle: $a = 5, b = 12, c = ?$

(6) Right triangle: $a = ?, b = 24, c = 25$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = 8, b = ?, c = 17$

(2) Right triangle: $a = 5, b = 12, c = ?$

(3) Right triangle: $a = ?, b = 4, c = 5$

(4) Right triangle: $a = 5, b = 12, c = ?$

(5) Right triangle: $a = ?, b = 8, c = 10$

(6) Right triangle: $a = ?, b = 24, c = 25$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = 7, b = ?, c = 25$

(2) Right triangle: $a = 6, b = ?, c = 10$

(3) Right triangle: $a = ?, b = 12, c = 13$

(4) Right triangle: $a = 6, b = ?, c = 10$

(5) Right triangle: $a = 3, b = ?, c = 5$

(6) Right triangle: $a = 7, b = 24, c = ?$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = 9, b = 12, c = ?$

(2) Right triangle: $a = 9, b = 12, c = ?$

(3) Right triangle: $a = 8, b = 15, c = ?$

(4) Right triangle: $a = 6, b = 8, c = ?$

(5) Right triangle: $a = ?, b = 8, c = 10$

(6) Right triangle: $a = ?, b = 12, c = 15$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = ?, b = 12, c = 13$

(2) Right triangle: $a = 5, b = ?, c = 13$

(3) Right triangle: $a = ?, b = 8, c = 10$

(4) Right triangle: $a = 9, b = 12, c = ?$

(5) Right triangle: $a = 9, b = ?, c = 15$

(6) Right triangle: $a = 7, b = 24, c = ?$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = 7, b = 24, c = ?$

(2) Right triangle: $a = 6, b = ?, c = 10$

(3) Right triangle: $a = 3, b = ?, c = 5$

(4) Right triangle: $a = ?, b = 12, c = 13$

(5) Right triangle: $a = 9, b = ?, c = 15$

(6) Right triangle: $a = 8, b = ?, c = 17$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = ?, b = 15, c = 17$

(2) Right triangle: $a = 6, b = 8, c = ?$

(3) Right triangle: $a = 7, b = 24, c = ?$

(4) Right triangle: $a = ?, b = 12, c = 15$

(5) Right triangle: $a = 3, b = 4, c = ?$

(6) Right triangle: $a = 8, b = ?, c = 17$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = 6, b = ?, c = 10$

(2) Right triangle: $a = 3, b = 4, c = ?$

(3) Right triangle: $a = 8, b = 15, c = ?$

(4) Right triangle: $a = ?, b = 12, c = 13$

(5) Right triangle: $a = ?, b = 8, c = 10$

(6) Right triangle: $a = ?, b = 8, c = 10$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = 5, b = 12, c = ?$

(2) Right triangle: $a = 6, b = 8, c = ?$

(3) Right triangle: $a = ?, b = 12, c = 13$

(4) Right triangle: $a = 9, b = ?, c = 15$

(5) Right triangle: $a = ?, b = 24, c = 25$

(6) Right triangle: $a = 7, b = ?, c = 25$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = ?, b = 12, c = 15$

(2) Right triangle: $a = 8, b = 15, c = ?$

(3) Right triangle: $a = ?, b = 4, c = 5$

(4) Right triangle: $a = 5, b = 12, c = ?$

(5) Right triangle: $a = ?, b = 8, c = 10$

(6) Right triangle: $a = ?, b = 12, c = 15$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = 5, b = 12, c = ?$

(2) Right triangle: $a = ?, b = 12, c = 15$

(3) Right triangle: $a = 9, b = ?, c = 15$

(4) Right triangle: $a = ?, b = 4, c = 5$

(5) Right triangle: $a = 6, b = 8, c = ?$

(6) Right triangle: $a = 9, b = ?, c = 15$

Section 10. The Pythagorean Theorem

Use the Pythagorean theorem to find the missing side.

(1) Right triangle: $a = 6, b = 8, c = ?$

(2) Right triangle: $a = ?, b = 15, c = 17$

(3) Right triangle: $a = 6, b = ?, c = 10$

(4) Right triangle: $a = 6, b = 8, c = ?$

(5) Right triangle: $a = 7, b = ?, c = 25$

(6) Right triangle: $a = 3, b = ?, c = 5$

Answer Key

JPMethod Polynomials

F1a

- (1) $2a^2 + 8a$
- (2) $2a^2 + 18a$
- (3) $x^2 + 2x + 1$
- (4) $3a^2 - 27a$
- (5) $5a^2 + 5a$
- (6) $y^2 + y - 6$

F2a

- (1) $3x^2 - 21x$
- (2) $4x^2 - 4x$
- (3) $y^2 + y - 2$
- (4) $y^2 - 2y - 8$
- (5) $4a^2 - 28a$
- (6) $x^2 + 3x - 10$

F1b

- (1) $2y^2 - 12y$
- (2) $x^2 - x - 20$
- (3) $4a^2 + 24a$
- (4) $2a^2 + 10a$
- (5) $2y^2 - 16y$
- (6) $4y^2 + 20y$

F2b

- (1) $2x^2 - 6x$
- (2) $5y^2 - 40y$
- (3) $2a^2 - 10a$
- (4) $4y^2 + 32y$
- (5) $x^2 + 5x - 6$
- (6) $4x^2 + 36x$

Answer Key

JPMethod Polynomials

F5a

- (1) $x^2 + 5x + 4$
- (2) $4x^2 + 8x$
- (3) $y^2 + 2y - 8$
- (4) $x^2 + 5x + 6$
- (5) $4y^2 - 16y$
- (6) $4a^2 + 20a$

F6a

- (1) $x^2 - 8x + 12$
- (2) $4x^2 - 8x$
- (3) $5x^2 - 40x$
- (4) $x^2 + x - 2$
- (5) $y^2 - 4y + 3$
- (6) $x^2 - 2x - 3$

F5b

- (1) $5y^2 - 5y$
- (2) $4x^2 - 32x$
- (3) $2x^2 + 6x$
- (4) $x^2 + 2x - 8$
- (5) $x^2 + 5x + 6$
- (6) $y^2 + 4y + 4$

F6b

- (1) $2a^2 - 12a$
- (2) $x^2 - x - 20$
- (3) $5x^2 + 45x$
- (4) $y^2 - 7y + 6$
- (5) $4a^2 - 28a$
- (6) $y^2 - 4y + 4$

Answer Key

JPMethod Polynomials

F7a

- (1) $4y^2 - 4y$
- (2) $x^2 - 9x + 20$
- (3) $y^2 + 3y - 10$
- (4) $4a^2 + 24a$
- (5) $4x^2 + 12x$
- (6) $3y^2 - 6y$

F8a

- (1) $4y^2 + 36y$
- (2) $3x^2 + 9x$
- (3) $5y^2 - 45y$
- (4) $x^2 + 2x - 8$
- (5) $y^2 - 4y + 3$
- (6) $x^2 - 6x + 8$

F7b

- (1) $x^2 - 16$
- (2) $2x^2 + 8x$
- (3) $x^2 - x - 2$
- (4) $y^2 - 9y + 20$
- (5) $5y^2 - 20y$
- (6) $x^2 - x - 12$

F8b

- (1) $x^2 - x - 12$
- (2) $x^2 + 5x + 4$
- (3) $x^2 + 2x - 8$
- (4) $y^2 - 6y + 8$
- (5) $y^2 + y - 12$
- (6) $2y^2 + 4y$

Answer Key

JPMethod Polynomials

F3a

- (1) $y^2 - 2y + 1$
- (2) $2x^2 + 4x$
- (3) $x^2 - 4$
- (4) $y^2 - 3y - 10$
- (5) $y^2 + y - 20$
- (6) $3x^2 + 3x$

F4a

- (1) $3x^2 + 24x$
- (2) $x^2 + 7x + 12$
- (3) $x^2 + 7x + 6$
- (4) $3y^2 + 24y$
- (5) $x^2 - 2x - 8$
- (6) $y^2 + y - 12$

F3b

- (1) $2x^2 + 2x$
- (2) $2y^2 + 18y$
- (3) $y^2 - 3y - 10$
- (4) $5y^2 + 10y$
- (5) $y^2 - 16$
- (6) $2y^2 + 12y$

F4b

- (1) $4x^2 + 36x$
- (2) $x^2 + 4x - 5$
- (3) $2x^2 + 4x$
- (4) $x^2 + 7x + 10$
- (5) $y^2 - y - 6$
- (6) $y^2 - 8y + 12$

Answer Key

JPMethod Polynomials

F9a

- (1) $3x^2 + 9x$
- (2) $5a^2 - 20a$
- (3) $y^2 + 7y + 10$
- (4) $x^2 - 4x + 3$
- (5) $2a^2 + 8a$
- (6) $x^2 - 4x + 3$

F10a

- (1) $y^2 - 9$
- (2) $2x^2 - 14x$
- (3) $5a^2 - 40a$
- (4) $2y^2 + 14y$
- (5) $y^2 - 9y + 20$
- (6) $y^2 - 7y + 12$

F9b

- (1) $4x^2 + 32x$
- (2) $y^2 - 2y - 24$
- (3) $4a^2 - 32a$
- (4) $y^2 + y - 6$
- (5) $x^2 - 9x + 20$
- (6) $5y^2 + 40y$

F10b

- (1) $2a^2 + 12a$
- (2) $3x^2 - 24x$
- (3) $x^2 - x - 6$
- (4) $4x^2 + 36x$
- (5) $x^2 + 4x - 5$
- (6) $x^2 - 6x + 5$

Answer Key

JPMethod Polynomials

F13a

- (1) $x^2 - x - 2$
- (2) $5a^2 - 45a$
- (3) $y^2 - 2y - 8$
- (4) $x^2 - 3x + 2$
- (5) $y^2 - 4y + 3$
- (6) $5a^2 - 30a$

F14a

- (1) $y^2 + 9y + 20$
- (2) $4x^2 - 8x$
- (3) $2y^2 + 10y$
- (4) $4x^2 - 24x$
- (5) $x^2 + 3x - 10$
- (6) $3x^2 + 21x$

F13b

- (1) $y^2 + y - 6$
- (2) $5x^2 - 40x$
- (3) $x^2 + 3x - 4$
- (4) $5x^2 + 40x$
- (5) $5a^2 - 10a$
- (6) $y^2 - 2y - 24$

F14b

- (1) $x^2 - 6x + 8$
- (2) $x^2 + 7x + 12$
- (3) $y^2 - 9y + 18$
- (4) $y^2 + 2y - 8$
- (5) $5a^2 - 25a$
- (6) $5y^2 - 5y$

Answer Key

JPMethod Polynomials

F15a

- (1) $x^2 + x - 6$
- (2) $2a^2 - 2a$
- (3) $x^2 + 9x + 18$
- (4) $y^2 + 6y + 8$
- (5) $y^2 - 5y + 6$
- (6) $x^2 + 5x + 4$

F16a

- (1) $y^2 - y - 20$
- (2) $5a^2 + 25a$
- (3) $x^2 - 16$
- (4) $4a^2 + 24a$
- (5) $y^2 - y - 2$
- (6) $2x^2 + 8x$

F15b

- (1) $y^2 - 3y - 4$
- (2) $5y^2 - 25y$
- (3) $x^2 - 2x - 8$
- (4) $2a^2 - 10a$
- (5) $3x^2 - 15x$
- (6) $2x^2 - 6x$

F16b

- (1) $2a^2 + 8a$
- (2) $x^2 + 4x + 3$
- (3) $x^2 + 7x + 6$
- (4) $4x^2 + 24x$
- (5) $4x^2 + 28x$
- (6) $5y^2 + 45y$

Answer Key

JPMethod Polynomials

F11a

- (1) $y^2 + y - 6$
- (2) $x^2 + 3x - 4$
- (3) $3x^2 + 15x$
- (4) $2a^2 - 8a$
- (5) $y^2 - 8y + 15$
- (6) $2x^2 - 8x$

F12a

- (1) $5a^2 - 35a$
- (2) $y^2 - 4y - 5$
- (3) $y^2 + 2y + 1$
- (4) $x^2 - 6x + 8$
- (5) $5a^2 - 10a$
- (6) $y^2 + 6y + 9$

F11b

- (1) $3x^2 + 27x$
- (2) $2y^2 - 16y$
- (3) $3y^2 - 15y$
- (4) $3y^2 - 12y$
- (5) $3a^2 + 15a$
- (6) $2x^2 - 10x$

F12b

- (1) $y^2 - y - 12$
- (2) $x^2 - 9$
- (3) $x^2 + 8x + 16$
- (4) $y^2 + 3y - 4$
- (5) $4y^2 + 32y$
- (6) $3a^2 + 15a$

F17a

- (1) $x^2 - 7x + 6$
- (2) $2y^2 + 14y$
- (3) $y^2 - 1$
- (4) $3y^2 - 27y$
- (5) $y^2 + y - 20$
- (6) $3y^2 - 12y$

F18a

- (1) $x^2 + 2x - 3$
- (2) $x^2 - x - 20$
- (3) $3y^2 - 15y$
- (4) $x^2 - 4$
- (5) $2x^2 + 18x$
- (6) $y^2 + y - 12$

F17b

- (1) $y^2 - 16$
- (2) $y^2 - 9$
- (3) $2x^2 + 14x$
- (4) $x^2 + 4x - 5$
- (5) $y^2 - 2y - 24$
- (6) $y^2 + 6y + 5$

F18b

- (1) $y^2 + 2y + 1$
- (2) $2y^2 - 14y$
- (3) $y^2 + y - 6$
- (4) $x^2 - 9x + 18$
- (5) $3y^2 + 3y$
- (6) $4x^2 - 20x$

F21a

- (1) $a^2 - 64$
- (2) $a^2 - 36$
- (3) $a^2 + 14a + 49$
- (4) $a^2 - 8a + 16$
- (5) $y^2 + 8y + 16$
- (6) $x^2 + 8x + 16$

F22a

- (1) $x^2 - 6x + 9$
- (2) $a^2 + 16a + 64$
- (3) $x^2 - 49$
- (4) $x^2 - 10x + 25$
- (5) $a^2 - 14a + 49$
- (6) $y^2 + 12y + 36$

F21b

- (1) $x^2 - 6x + 9$
- (2) $x^2 + 4x + 4$
- (3) $x^2 - 12x + 36$
- (4) $a^2 - 36$
- (5) $a^2 + 8a + 16$
- (6) $x^2 - 36$

F22b

- (1) $a^2 - 16a + 64$
- (2) $x^2 - 12x + 36$
- (3) $y^2 + 16y + 64$
- (4) $x^2 - 8x + 16$
- (5) $a^2 - 25$
- (6) $y^2 - 16y + 64$

Answer Key

JPMethod Polynomials

F23a

- (1) $a^2 - 12a + 36$
- (2) $x^2 - 16x + 64$
- (3) $a^2 + 6a + 9$
- (4) $x^2 - 14x + 49$
- (5) $y^2 - 12y + 36$
- (6) $a^2 + 16a + 64$

F24a

- (1) $y^2 - 9$
- (2) $x^2 - 8x + 16$
- (3) $x^2 + 8x + 16$
- (4) $a^2 + 14a + 49$
- (5) $x^2 - 6x + 9$
- (6) $y^2 + 16y + 64$

F23b

- (1) $x^2 - 16$
- (2) $x^2 + 8x + 16$
- (3) $a^2 - 6a + 9$
- (4) $x^2 - 6x + 9$
- (5) $y^2 - 4y + 4$
- (6) $x^2 - 10x + 25$

F24b

- (1) $a^2 - 16a + 64$
- (2) $a^2 + 10a + 25$
- (3) $a^2 - 4a + 4$
- (4) $x^2 + 8x + 16$
- (5) $x^2 + 12x + 36$
- (6) $a^2 - 36$

Answer Key

JPMethod Polynomials

F19a

- (1) $x^2 + 3x - 4$
- (2) $5a^2 - 20a$
- (3) $5a^2 + 35a$
- (4) $x^2 - 8x + 15$
- (5) $y^2 - 3y - 4$
- (6) $5a^2 - 10a$

F20a

- (1) $5x^2 - 40x$
- (2) $x^2 - 2x - 15$
- (3) $y^2 - 2y + 1$
- (4) $2a^2 - 18a$
- (5) $x^2 - x - 20$
- (6) $x^2 - 3x - 4$

F19b

- (1) $3y^2 + 21y$
- (2) $y^2 - y - 6$
- (3) $2a^2 - 8a$
- (4) $x^2 + x - 2$
- (5) $3a^2 + 6a$
- (6) $y^2 - 10y + 24$

F20b

- (1) $3x^2 - 24x$
- (2) $4y^2 - 28y$
- (3) $x^2 - 2x - 3$
- (4) $y^2 - 3y - 18$
- (5) $3y^2 + 21y$
- (6) $4a^2 + 24a$

Answer Key

JPMethod Polynomials

F25a

- (1) $x^2 - 16x + 64$
- (2) $x^2 - 8x + 16$
- (3) $x^2 + 6x + 9$
- (4) $a^2 - 16a + 64$
- (5) $y^2 - 4$
- (6) $y^2 + 12y + 36$

F26a

- (1) $x^2 - 36$
- (2) $x^2 - 12x + 36$
- (3) $a^2 - 64$
- (4) $a^2 - 4$
- (5) $y^2 - 4y + 4$
- (6) $y^2 - 16$

F25b

- (1) $x^2 - 49$
- (2) $a^2 - 9$
- (3) $x^2 - 16$
- (4) $x^2 - 12x + 36$
- (5) $x^2 - 12x + 36$
- (6) $x^2 - 14x + 49$

F26b

- (1) $a^2 - 4a + 4$
- (2) $x^2 - 8x + 16$
- (3) $a^2 + 14a + 49$
- (4) $x^2 - 64$
- (5) $a^2 - 64$
- (6) $x^2 - 4x + 4$

Answer Key

JPMethod Polynomials

F29a

- (1) $y^2 - 12y + 36$
- (2) $y^2 - 9$
- (3) $y^2 - 16$
- (4) $x^2 - 14x + 49$
- (5) $a^2 - 10a + 25$
- (6) $a^2 - 16a + 64$

F30a

- (1) $x^2 + 16x + 64$
- (2) $y^2 - 16$
- (3) $a^2 - 4a + 4$
- (4) $x^2 - 49$
- (5) $y^2 - 8y + 16$
- (6) $x^2 - 12x + 36$

F29b

- (1) $x^2 + 14x + 49$
- (2) $y^2 - 16$
- (3) $a^2 - 49$
- (4) $x^2 + 6x + 9$
- (5) $a^2 - 64$
- (6) $x^2 - 6x + 9$

F30b

- (1) $a^2 + 8a + 16$
- (2) $y^2 - 14y + 49$
- (3) $x^2 - 4x + 4$
- (4) $a^2 + 6a + 9$
- (5) $a^2 - 10a + 25$
- (6) $x^2 - 16$

Answer Key

JPMethod Polynomials

F31a

- (1) $(y + 5)(y - 5)$
- (2) $(y + 4)(y + 2)$
- (3) $3x(8x + 2)$
- (4) $4y(2y + 4)$
- (5) $(x + 6)(x - 6)$
- (6) $(x + 4)(x - 4)$

F32a

- (1) $3x(4x + 2)$
- (2) $(y + 4)(y - 4)$
- (3) $(x + 2)(x + 4)$
- (4) $(x + 1)(x + 5)$
- (5) $3y(3y + 3)$
- (6) $(x + 3)(x + 5)$

F31b

- (1) $(x + 3)(x - 3)$
- (2) $2x(3x + 5)$
- (3) $(y + 9)(y - 9)$
- (4) $(x + 1)(x + 1)$
- (5) $(y + 9)(y - 9)$
- (6) $(y + 4)(y + 5)$

F32b

- (1) $(x + 6)(x - 6)$
- (2) $4y(3y + 4)$
- (3) $(x + 2)(x + 5)$
- (4) $(x + 2)(x + 6)$
- (5) $(y + 1)(y + 1)$
- (6) $3x(7x + 6)$

Answer Key

JPMethod Polynomials

F27a

- (1) $a^2 - 8a + 16$
- (2) $x^2 - 64$
- (3) $y^2 - 49$
- (4) $a^2 - 6a + 9$
- (5) $x^2 + 14x + 49$
- (6) $x^2 - 8x + 16$

F28a

- (1) $a^2 + 8a + 16$
- (2) $x^2 - 16x + 64$
- (3) $y^2 - 10y + 25$
- (4) $y^2 - 25$
- (5) $x^2 - 10x + 25$
- (6) $y^2 - 16$

F27b

- (1) $x^2 + 16x + 64$
- (2) $x^2 + 8x + 16$
- (3) $a^2 - 10a + 25$
- (4) $y^2 - 16$
- (5) $a^2 + 6a + 9$
- (6) $y^2 - 16y + 64$

F28b

- (1) $y^2 - 64$
- (2) $a^2 + 10a + 25$
- (3) $y^2 + 12y + 36$
- (4) $a^2 + 8a + 16$
- (5) $x^2 - 4x + 4$
- (6) $a^2 - 8a + 16$

Answer Key

JPMethod Polynomials

F33a

- (1) $(x + 4)(x + 5)$
- (2) $(x + 4)(x + 6)$
- (3) $(y + 3)(y + 4)$
- (4) $2y(4y + 5)$
- (5) $(x + 2)(x + 4)$
- (6) $(y + 9)(y - 9)$

F34a

- (1) $(x + 4)(x - 4)$
- (2) $(y + 5)(y - 5)$
- (3) $3y(6y + 5)$
- (4) $(y + 2)(y - 2)$
- (5) $2y(6y + 5)$
- (6) $3y(2y + 2)$

F33b

- (1) $3y(6y + 3)$
- (2) $(y + 3)(y + 4)$
- (3) $2y(6y + 6)$
- (4) $(x + 9)(x - 9)$
- (5) $2y(5y + 3)$
- (6) $4x(4x + 5)$

F34b

- (1) $(y + 6)(y - 6)$
- (2) $(x + 1)(x + 4)$
- (3) $3y(7y + 4)$
- (4) $(x + 8)(x - 8)$
- (5) $3x(4x + 6)$
- (6) $4y(6y + 5)$

Answer Key

JPMethod Polynomials

F37a

- (1) $4y(8y + 4)$
- (2) $3y(4y + 5)$
- (3) $(y + 3)(y + 1)$
- (4) $(y + 4)(y + 3)$
- (5) $4y(2y + 4)$
- (6) $4y(5y + 4)$

F38a

- (1) $3y(7y + 5)$
- (2) $(x + 3)(x + 3)$
- (3) $(x + 3)(x - 3)$
- (4) $(x + 1)(x + 2)$
- (5) $(x + 2)(x - 2)$
- (6) $(y + 4)(y - 4)$

F37b

- (1) $(x + 5)(x - 5)$
- (2) $(y + 3)(y + 2)$
- (3) $4x(3x + 2)$
- (4) $(x + 2)(x + 1)$
- (5) $2y(4y + 5)$
- (6) $(x + 1)(x + 5)$

F38b

- (1) $(y + 1)(y + 1)$
- (2) $(x + 7)(x - 7)$
- (3) $(y + 9)(y - 9)$
- (4) $2y(4y + 3)$
- (5) $(x + 9)(x - 9)$
- (6) $(y + 2)(y + 1)$

Answer Key

JPMethod Polynomials

F39a

- (1) $(y + 3)(y + 5)$
- (2) $(x + 2)(x + 5)$
- (3) $(y + 4)(y + 5)$
- (4) $(x + 8)(x - 8)$
- (5) $(x + 2)(x + 2)$
- (6) $(y + 5)(y - 5)$

F40a

- (1) $(x + 6)(x - 6)$
- (2) $(x + 5)(x - 5)$
- (3) $4y(3y + 5)$
- (4) $(y + 4)(y + 6)$
- (5) $(y + 2)(y + 2)$
- (6) $(x + 4)(x + 5)$

F39b

- (1) $3y(5y + 3)$
- (2) $(y + 6)(y - 6)$
- (3) $(x + 3)(x + 6)$
- (4) $(x + 2)(x - 2)$
- (5) $(x + 4)(x + 3)$
- (6) $4y(7y + 6)$

F40b

- (1) $(y + 1)(y + 5)$
- (2) $4x(8x + 6)$
- (3) $4x(3x + 4)$
- (4) $(x + 5)(x - 5)$
- (5) $(y + 1)(y + 5)$
- (6) $(x + 9)(x - 9)$

Answer Key

JPMethod Polynomials

F35a

- (1) $(x + 4)(x - 4)$
- (2) $4y(4y + 4)$
- (3) $(x + 5)(x - 5)$
- (4) $(y + 1)(y + 1)$
- (5) $3x(7x + 4)$
- (6) $(y + 2)(y + 5)$

F36a

- (1) $(y + 5)(y - 5)$
- (2) $(x + 4)(x + 2)$
- (3) $(y + 2)(y + 6)$
- (4) $2x(6x + 2)$
- (5) $2y(4y + 5)$
- (6) $(x + 8)(x - 8)$

F35b

- (1) $(x + 3)(x + 5)$
- (2) $(x + 8)(x - 8)$
- (3) $(y + 3)(y + 5)$
- (4) $(y + 3)(y + 4)$
- (5) $3y(4y + 2)$
- (6) $(y + 1)(y + 5)$

F36b

- (1) $(y + 1)(y + 6)$
- (2) $4x(6x + 4)$
- (3) $(y + 7)(y - 7)$
- (4) $4y(6y + 4)$
- (5) $(y + 9)(y - 9)$
- (6) $3x(4x + 6)$

Answer Key

JPMethod Polynomials

F41a

- (1) $2y(5y + 6)$
- (2) $(y + 3)(y + 6)$
- (3) $4y(4y + 2)$
- (4) $3x(2x + 5)$
- (5) $(x + 1)(x + 1)$
- (6) $(x + 6)(x - 6)$

F42a

- (1) $(y + 8)(y - 8)$
- (2) $2x(5x + 4)$
- (3) $(x + 3)(x - 3)$
- (4) $2y(6y + 2)$
- (5) $2x(2x + 6)$
- (6) $(x + 5)(x - 5)$

F41b

- (1) $(x + 4)(x - 4)$
- (2) $(y + 3)(y + 4)$
- (3) $(y + 8)(y - 8)$
- (4) $2x(7x + 4)$
- (5) $4y(6y + 3)$
- (6) $3x(8x + 2)$

F42b

- (1) $(x + 4)(x + 4)$
- (2) $2x(8x + 2)$
- (3) $(y + 2)(y + 6)$
- (4) $(y + 3)(y + 2)$
- (5) $(x + 5)(x - 5)$
- (6) $(y + 6)(y - 6)$

Answer Key

JPMethod Polynomials

F45a

- (1) $(y + 6)(y - 6)$
- (2) $3x(4x + 3)$
- (3) $(x + 4)(x + 5)$
- (4) $(y + 3)(y + 4)$
- (5) $(y + 4)(y + 2)$
- (6) $(y + 2)(y + 2)$

F46a

- (1) $4y(2y + 2)$
- (2) $(y + 3)(y + 1)$
- (3) $(y + 2)(y + 1)$
- (4) $4x(3x + 3)$
- (5) $(y + 3)(y + 3)$
- (6) $(x + 7)(x - 7)$

F45b

- (1) $(y + 2)(y + 5)$
- (2) $(x + 1)(x + 1)$
- (3) $(x + 1)(x + 4)$
- (4) $2x(2x + 3)$
- (5) $4y(4y + 2)$
- (6) $(y + 4)(y + 1)$

F46b

- (1) $(y + 5)(y - 5)$
- (2) $(x + 3)(x + 6)$
- (3) $3x(2x + 5)$
- (4) $(y + 2)(y - 2)$
- (5) $(x + 2)(x + 1)$
- (6) $(y + 7)(y - 7)$

F47a

- (1) $(x + 1)(x + 2)$
- (2) $2y(4y + 2)$
- (3) $(y + 1)(y + 2)$
- (4) $(x + 5)(x - 5)$
- (5) $(x + 4)(x + 1)$
- (6) $(x + 5)(x - 5)$

F48a

- (1) $3y(6y + 5)$
- (2) $3y(2y + 6)$
- (3) $2x(2x + 4)$
- (4) $(x + 8)(x - 8)$
- (5) $(y + 3)(y - 3)$
- (6) $3x(4x + 5)$

F47b

- (1) $(y + 3)(y + 4)$
- (2) $(x + 1)(x + 5)$
- (3) $(y + 1)(y + 1)$
- (4) $(x + 3)(x - 3)$
- (5) $(y + 7)(y - 7)$
- (6) $3y(8y + 5)$

F48b

- (1) $(x + 4)(x - 4)$
- (2) $4x(5x + 5)$
- (3) $(y + 1)(y + 1)$
- (4) $4y(6y + 2)$
- (5) $(y + 4)(y + 1)$
- (6) $(y + 8)(y - 8)$

F43a

- (1) $4x(7x + 5)$
- (2) $(x + 7)(x - 7)$
- (3) $(y + 3)(y + 6)$
- (4) $(y + 4)(y - 4)$
- (5) $(y + 4)(y - 4)$
- (6) $(y + 2)(y + 2)$

F44a

- (1) $4x(3x + 6)$
- (2) $(x + 4)(x + 2)$
- (3) $(y + 2)(y - 2)$
- (4) $4x(7x + 2)$
- (5) $(x + 5)(x - 5)$
- (6) $(x + 7)(x - 7)$

F43b

- (1) $(x + 9)(x - 9)$
- (2) $4y(5y + 6)$
- (3) $2y(3y + 4)$
- (4) $(x + 9)(x - 9)$
- (5) $(x + 5)(x - 5)$
- (6) $4y(5y + 6)$

F44b

- (1) $(y + 9)(y - 9)$
- (2) $(y + 3)(y + 3)$
- (3) $(y + 9)(y - 9)$
- (4) $2y(8y + 4)$
- (5) $(y + 8)(y - 8)$
- (6) $2x(7x + 3)$

Answer Key

JPMethod Polynomials

F49a

- (1) $2x(6x + 6)$
- (2) $(y + 2)(y + 2)$
- (3) $(y + 6)(y - 6)$
- (4) $(x + 3)(x - 3)$
- (5) $3y(4y + 5)$
- (6) $(x + 7)(x - 7)$

F50a

- (1) $(y + 4)(y - 4)$
- (2) $(y + 6)(y - 6)$
- (3) $(y + 2)(y + 3)$
- (4) $(y + 8)(y - 8)$
- (5) $(x + 5)(x - 5)$
- (6) $(x + 1)(x + 6)$

F49b

- (1) $(x + 9)(x - 9)$
- (2) $(x + 4)(x + 1)$
- (3) $3y(2y + 5)$
- (4) $(y + 3)(y + 4)$
- (5) $(y + 4)(y - 4)$
- (6) $(y + 5)(y - 5)$

F50b

- (1) $(x + 9)(x - 9)$
- (2) $(x + 8)(x - 8)$
- (3) $(y + 2)(y + 6)$
- (4) $(x + 6)(x - 6)$
- (5) $(x + 4)(x + 4)$
- (6) $4y(6y + 5)$

Answer Key

JPMethod Polynomials

F53a

- (1) $5\sqrt{3}$
- (2) $6\sqrt{3}$
- (3) $5\sqrt{7}$
- (4) $5\sqrt{2}$
- (5) 8
- (6) $4\sqrt{6}$

F54a

- (1) $5\sqrt{5}$
- (2) 9
- (3) 6
- (4) $8\sqrt{5}$
- (5) $8\sqrt{6}$
- (6) $4\sqrt{5}$

F53b

- (1) $2\sqrt{2}$
- (2) $5\sqrt{7}$
- (3) $3\sqrt{5}$
- (4) 5
- (5) 6
- (6) $5\sqrt{2}$

F54b

- (1) $3\sqrt{2}$
- (2) $7\sqrt{7}$
- (3) 4
- (4) $2\sqrt{3}$
- (5) 6
- (6) 4

Answer Key

JPMethod Polynomials

F55a

- (1) $3\sqrt{2}$
- (2) $7\sqrt{7}$
- (3) 8
- (4) $3\sqrt{2}$
- (5) $3\sqrt{2}$
- (6) $5\sqrt{5}$

F56a

- (1) $5\sqrt{2}$
- (2) $3\sqrt{2}$
- (3) 2
- (4) 6
- (5) 6
- (6) 10

F55b

- (1) 4
- (2) $5\sqrt{2}$
- (3) $4\sqrt{2}$
- (4) $3\sqrt{2}$
- (5) 8
- (6) $7\sqrt{6}$

F56b

- (1) $3\sqrt{2}$
- (2) $5\sqrt{2}$
- (3) $5\sqrt{2}$
- (4) $4\sqrt{2}$
- (5) $5\sqrt{6}$
- (6) $5\sqrt{2}$

Answer Key

JPMethod Polynomials

F51a

- (1) $3\sqrt{5}$
- (2) $4\sqrt{7}$
- (3) $3\sqrt{3}$
- (4) $5\sqrt{7}$
- (5) 8
- (6) $3\sqrt{5}$

F52a

- (1) $2\sqrt{3}$
- (2) $3\sqrt{5}$
- (3) 3
- (4) $5\sqrt{5}$
- (5) $5\sqrt{2}$
- (6) 10

F51b

- (1) 2
- (2) 6
- (3) 10
- (4) $5\sqrt{5}$
- (5) $5\sqrt{2}$
- (6) $5\sqrt{2}$

F52b

- (1) $4\sqrt{2}$
- (2) $2\sqrt{3}$
- (3) $5\sqrt{2}$
- (4) $2\sqrt{3}$
- (5) $2\sqrt{2}$
- (6) $3\sqrt{5}$

F57a

- (1) $5\sqrt{2}$
- (2) $3\sqrt{3}$
- (3) $2\sqrt{3}$
- (4) $2\sqrt{3}$
- (5) 5
- (6) $7\sqrt{7}$

F58a

- (1) $3\sqrt{6}$
- (2) 6
- (3) $6\sqrt{3}$
- (4) 4
- (5) $5\sqrt{6}$
- (6) $3\sqrt{2}$

F57b

- (1) $6\sqrt{7}$
- (2) 5
- (3) 2
- (4) $7\sqrt{5}$
- (5) $5\sqrt{5}$
- (6) $3\sqrt{3}$

F58b

- (1) $7\sqrt{6}$
- (2) $5\sqrt{2}$
- (3) 3
- (4) $3\sqrt{3}$
- (5) $6\sqrt{6}$
- (6) $2\sqrt{2}$

F61a

- (1) $y = \pm 1$
- (2) $y = \pm 2$
- (3) $x = \pm 1$
- (4) $y = -1, -1$
- (5) $y = -3, -3$
- (6) $y = -2, -2$

F62a

- (1) $x = \pm 2$
- (2) $x = \pm 2$
- (3) $x = -4, -3$
- (4) $x = \pm 2$
- (5) $(x + 3)^2 = 17$
- (6) $x = \pm 1$

F61b

- (1) $x = -2, -4$
- (2) $(x + 7)^2 = 52$
- (3) $(y + 8)^2 = 68$
- (4) $x = -2, -1$
- (5) $y = \pm 2$
- (6) $(x + 3)^2 = 14$

F62b

- (1) $(y + 3)^2 = 16$
- (2) $y = \pm 1$
- (3) $(x + 2)^2 = 14$
- (4) $x = -3, -2$
- (5) $x = -4, -3$
- (6) $y = \pm 2$

Answer Key

JPMethod Polynomials

F63a

- (1) $x = -3, -1$
- (2) $(x + 6)^2 = 40$
- (3) $y = -4, -1$
- (4) $(y + 7)^2 = 53$
- (5) $x = -1, -4$
- (6) $y = -1, -3$

F64a

- (1) $(x + 5)^2 = 33$
- (2) $y = \pm 2$
- (3) $x = \pm 1$
- (4) $(x + 6)^2 = 38$
- (5) $x = -1, -1$
- (6) $x = -4, -3$

F63b

- (1) $x = \pm 1$
- (2) $x = \pm 1$
- (3) $y = -3, -3$
- (4) $y = -2, -3$
- (5) $x = \pm 2$
- (6) $(y + 6)^2 = 38$

F64b

- (1) $(x + 3)^2 = 10$
- (2) $y = \pm 2$
- (3) $y = -3, -3$
- (4) $y = -1, -1$
- (5) $x = -1, -4$
- (6) $y = -3, -3$

Answer Key

JPMethod Polynomials

F59a

- (1) $5\sqrt{6}$
- (2) $5\sqrt{7}$
- (3) $2\sqrt{5}$
- (4) $2\sqrt{5}$
- (5) $3\sqrt{3}$
- (6) $3\sqrt{3}$

F60a

- (1) $2\sqrt{5}$
- (2) $6\sqrt{5}$
- (3) 6
- (4) 7
- (5) $2\sqrt{3}$
- (6) $3\sqrt{2}$

F59b

- (1) $3\sqrt{2}$
- (2) $7\sqrt{2}$
- (3) 3
- (4) $6\sqrt{2}$
- (5) 9
- (6) $2\sqrt{2}$

F60b

- (1) 8
- (2) $7\sqrt{3}$
- (3) $2\sqrt{3}$
- (4) $3\sqrt{3}$
- (5) $2\sqrt{3}$
- (6) 5

F65a

- (1) $(x + 6)^2 = 44$
- (2) $y = -2, -3$
- (3) $y = \pm 2$
- (4) $(y + 6)^2 = 44$
- (5) $x = -2, -1$
- (6) $(x + 3)^2 = 18$

F66a

- (1) $x = \pm 1$
- (2) $y = \pm 1$
- (3) $(x + 3)^2 = 14$
- (4) $(y + 8)^2 = 71$
- (5) $y = \pm 2$
- (6) $(x + 2)^2 = 8$

F65b

- (1) $(x + 4)^2 = 17$
- (2) $y = \pm 2$
- (3) $x = -2, -4$
- (4) $(x + 3)^2 = 14$
- (5) $x = -2, -4$
- (6) $y = \pm 1$

F66b

- (1) $x = -2, -1$
- (2) $y = -4, -3$
- (3) $(x + 6)^2 = 42$
- (4) $y = -3, -4$
- (5) $(x + 2)^2 = 7$
- (6) $y = \pm 1$

F69a

- (1) $y = -1, -4$
- (2) $y = \pm 2$
- (3) $x = -2, -2$
- (4) $(x + 8)^2 = 68$
- (5) $y = \pm 1$
- (6) $x = -3, -3$

F70a

- (1) $(x + 8)^2 = 70$
- (2) $x = -2, -2$
- (3) $y = -1, -3$
- (4) $y = \pm 1$
- (5) $(y + 4)^2 = 19$
- (6) $y = -4, -3$

F69b

- (1) $x = \pm 2$
- (2) $x = \pm 2$
- (3) $(y + 3)^2 = 17$
- (4) $x = -1, -2$
- (5) $x = \pm 2$
- (6) $y = -2, -2$

F70b

- (1) $y = -2, -2$
- (2) $(y + 7)^2 = 54$
- (3) $x = -1, -4$
- (4) $x = \pm 2$
- (5) $x = \pm 1$
- (6) $(x + 4)^2 = 18$

F71a

- (1) $y = -3, -2$
- (2) $x = \pm 2$
- (3) $(x + 4)^2 = 20$
- (4) $x = -3, -3$
- (5) $x = -1, -4$
- (6) $(y + 3)^2 = 16$

F72a

- (1) $x = \pm 1$
- (2) $(y + 7)^2 = 50$
- (3) $(x + 6)^2 = 43$
- (4) $x = -1, -3$
- (5) $x = -4, -3$
- (6) $y = \pm 2$

F71b

- (1) $(y + 4)^2 = 25$
- (2) $(x + 8)^2 = 72$
- (3) $x = \pm 2$
- (4) $x = -3, -1$
- (5) $(y + 2)^2 = 6$
- (6) $(x + 5)^2 = 28$

F72b

- (1) $x = -4, -4$
- (2) $x = -3, -1$
- (3) $(x + 4)^2 = 26$
- (4) $(x + 2)^2 = 5$
- (5) $x = -3, -1$
- (6) $(x + 4)^2 = 23$

F67a

- (1) $y = -2, -1$
- (2) $x = \pm 2$
- (3) $x = \pm 2$
- (4) $(x + 2)^2 = 11$
- (5) $(x + 2)^2 = 12$
- (6) $x = -1, -1$

F68a

- (1) $(y + 5)^2 = 26$
- (2) $(y + 3)^2 = 15$
- (3) $x = -3, -3$
- (4) $(x + 2)^2 = 14$
- (5) $x = \pm 2$
- (6) $(y + 6)^2 = 44$

F67b

- (1) $x = -2, -2$
- (2) $(y + 7)^2 = 51$
- (3) $y = \pm 1$
- (4) $x = -4, -2$
- (5) $y = -1, -1$
- (6) $(x + 3)^2 = 19$

F68b

- (1) $x = -3, -4$
- (2) $y = -2, -2$
- (3) $x = \pm 2$
- (4) $(y + 4)^2 = 26$
- (5) $y = \pm 2$
- (6) $x = -3, -2$

F73a

- (1) $(y + 7)^2 = 54$
- (2) $y = -2, -3$
- (3) $(y + 3)^2 = 17$
- (4) $y = -4, -4$
- (5) $(x + 8)^2 = 69$
- (6) $x = \pm 1$

F74a

- (1) $x = -2, -1$
- (2) $(x + 7)^2 = 56$
- (3) $x = -3, -3$
- (4) $x = \pm 2$
- (5) $y = \pm 2$
- (6) $(y + 8)^2 = 65$

F73b

- (1) $y = -4, -2$
- (2) $x = \pm 2$
- (3) $y = \pm 2$
- (4) $(y + 2)^2 = 11$
- (5) $x = \pm 1$
- (6) $y = -2, -2$

F74b

- (1) $y = \pm 2$
- (2) $x = -1, -1$
- (3) $x = \pm 2$
- (4) $(y + 5)^2 = 33$
- (5) $y = \pm 1$
- (6) $(y + 3)^2 = 19$

F77a

- (1) $y = -3, -3$
- (2) $(y + 8)^2 = 70$
- (3) $(y + 3)^2 = 17$
- (4) $(y + 6)^2 = 38$
- (5) $y = -3, -3$
- (6) $x = \pm 1$

F78a

- (1) $(y + 5)^2 = 26$
- (2) $x = \pm 2$
- (3) $y = \pm 2$
- (4) $y = \pm 1$
- (5) $(x + 5)^2 = 28$
- (6) $y = \pm 1$

F77b

- (1) $(y + 7)^2 = 55$
- (2) $(y + 7)^2 = 54$
- (3) $x = -2, -1$
- (4) $y = \pm 2$
- (5) $x = \pm 1$
- (6) $y = \pm 2$

F78b

- (1) $y = \pm 1$
- (2) $y = -2, -3$
- (3) $(y + 4)^2 = 21$
- (4) $y = \pm 1$
- (5) $(x + 8)^2 = 74$
- (6) $x = \pm 2$

Answer Key

JPMethod Polynomials

F79a

- (1) $x = \pm 2$
- (2) $y = \pm 1$
- (3) $x = -2, -3$
- (4) $(y + 3)^2 = 15$
- (5) $x = \pm 2$
- (6) $(x + 4)^2 = 17$

F80a

- (1) $x = \pm 2$
- (2) $(x + 2)^2 = 11$
- (3) $y = \pm 1$
- (4) $y = -3, -4$
- (5) $y = \pm 1$
- (6) $(y + 4)^2 = 20$

F79b

- (1) $y = -3, -2$
- (2) $x = -4, -1$
- (3) $y = -4, -4$
- (4) $(y + 5)^2 = 26$
- (5) $x = -3, -1$
- (6) $(x + 2)^2 = 5$

F80b

- (1) $y = -1, -2$
- (2) $x = -2, -1$
- (3) $x = \pm 2$
- (4) $x = \pm 2$
- (5) $x = \pm 1$
- (6) $x = \pm 1$

Answer Key

JPMethod Polynomials

F75a

- (1) $y = \pm 2$
- (2) $(x + 2)^2 = 12$
- (3) $(y + 2)^2 = 10$
- (4) $(y + 4)^2 = 24$
- (5) $x = \pm 1$
- (6) $(y + 8)^2 = 70$

F76a

- (1) $x = -4, -1$
- (2) $(x + 4)^2 = 22$
- (3) $(y + 7)^2 = 51$
- (4) $x = \pm 2$
- (5) $(y + 3)^2 = 17$
- (6) $y = \pm 1$

F75b

- (1) $(y + 6)^2 = 38$
- (2) $y = -2, -4$
- (3) $y = \pm 2$
- (4) $x = -1, -3$
- (5) $y = \pm 1$
- (6) $x = -2, -1$

F76b

- (1) $(x + 4)^2 = 17$
- (2) $(x + 3)^2 = 19$
- (3) $x = -2, -2$
- (4) $(y + 6)^2 = 37$
- (5) $x = \pm 1$
- (6) $x = \pm 1$

F81a

- (1) $(-7, -47)$
- (2) $x = \pm 5$
- (3) $x = \pm 2$
- (4) down
- (5) $x = \pm 3$
- (6) $(-7, -48)$

F82a

- (1) $(-4, -14)$
- (2) $(-5, -21)$
- (3) down
- (4) $x = \pm 3$
- (5) $x = \pm 5$
- (6) $x = \pm 3$

F81b

- (1) $x = \pm 5$
- (2) down
- (3) up
- (4) $x = \pm 3$
- (5) $x = \pm 2$
- (6) up

F82b

- (1) $(-6, -28)$
- (2) $(-7, -41)$
- (3) $x = \pm 3$
- (4) $(-2, -2)$
- (5) $(-8, -56)$
- (6) $(-2, 5)$

F85a

- (1) $(-6, -29)$
- (2) $x = \pm 2$
- (3) down
- (4) $x = \pm 5$
- (5) $(-5, -17)$
- (6) $(-2, 1)$

F86a

- (1) up
- (2) $(-5, -21)$
- (3) $x = \pm 4$
- (4) up
- (5) $(-8, -55)$
- (6) down

F85b

- (1) down
- (2) $(-7, -40)$
- (3) up
- (4) up
- (5) $(-7, -43)$
- (6) up

F86b

- (1) $(-7, -46)$
- (2) up
- (3) $x = \pm 3$
- (4) $(-8, -56)$
- (5) $x = \pm 3$
- (6) $x = \pm 4$

F87a

- (1) $(-6, -32)$
- (2) up
- (3) $x = \pm 3$
- (4) $(-5, -17)$
- (5) up
- (6) $(-8, -56)$

F88a

- (1) down
- (2) up
- (3) $(-7, -41)$
- (4) $x = \pm 4$
- (5) $(-8, -60)$
- (6) $x = \pm 4$

F87b

- (1) $x = \pm 4$
- (2) $(-5, -24)$
- (3) $(-5, -18)$
- (4) $(-6, -27)$
- (5) $(-8, -62)$
- (6) up

F88b

- (1) up
- (2) up
- (3) down
- (4) up
- (5) $(-2, 0)$
- (6) $(-4, -10)$

F83a

- (1) $(-7, -47)$
- (2) $x = \pm 3$
- (3) $x = \pm 4$
- (4) $x = \pm 5$
- (5) $(-2, -1)$
- (6) $(-8, -59)$

F84a

- (1) $x = \pm 4$
- (2) down
- (3) $x = \pm 4$
- (4) $(-2, 3)$
- (5) $x = \pm 5$
- (6) $x = \pm 5$

F83b

- (1) $x = \pm 4$
- (2) $x = \pm 2$
- (3) $(-8, -63)$
- (4) $(-8, -61)$
- (5) down
- (6) $x = \pm 4$

F84b

- (1) up
- (2) $x = \pm 3$
- (3) $(-3, -8)$
- (4) $(-2, 3)$
- (5) up
- (6) up

Answer Key

JPMethod Polynomials

F89a

- (1) down
- (2) up
- (3) $x = \pm 2$
- (4) $(-3, -1)$
- (5) $(-8, -57)$
- (6) $(-4, -12)$

F90a

- (1) up
- (2) down
- (3) $x = \pm 2$
- (4) down
- (5) up
- (6) up

F89b

- (1) $(-2, 3)$
- (2) $(-5, -21)$
- (3) down
- (4) $(-8, -57)$
- (5) $(-7, -47)$
- (6) $x = \pm 5$

F90b

- (1) down
- (2) up
- (3) $(-7, -46)$
- (4) $(-5, -19)$
- (5) up
- (6) $x = \pm 3$

Answer Key

JPMethod Polynomials

F93a

- (1) 15
- (2) 8
- (3) 7
- (4) 6
- (5) 4
- (6) 5

F94a

- (1) 13
- (2) 4
- (3) 7
- (4) 10
- (5) 15
- (6) 13

F93b

- (1) 9
- (2) 12
- (3) 12
- (4) 12
- (5) 5
- (6) 12

F94b

- (1) 15
- (2) 7
- (3) 10
- (4) 17
- (5) 9
- (6) 24

Answer Key

JPMethod Polynomials

F95a

- (1) 25
- (2) 17
- (3) 3
- (4) 12
- (5) 13
- (6) 7

F96a

- (1) 15
- (2) 13
- (3) 3
- (4) 13
- (5) 6
- (6) 7

F95b

- (1) 15
- (2) 15
- (3) 17
- (4) 10
- (5) 6
- (6) 9

F96b

- (1) 24
- (2) 8
- (3) 5
- (4) 8
- (5) 4
- (6) 25

Answer Key

JPMethod Polynomials

F91a

- (1) 15
- (2) 15
- (3) 7
- (4) 12
- (5) 10
- (6) 15

F92a

- (1) 12
- (2) 12
- (3) 9
- (4) 15
- (5) 13
- (6) 10

F91b

- (1) 17
- (2) 8
- (3) 15
- (4) 10
- (5) 5
- (6) 13

F92b

- (1) 12
- (2) 9
- (3) 17
- (4) 25
- (5) 10
- (6) 12

F97a

- (1) 5
- (2) 12
- (3) 6
- (4) 15
- (5) 12
- (6) 25

F98a

- (1) 25
- (2) 8
- (3) 4
- (4) 5
- (5) 12
- (6) 15

F97b

- (1) 8
- (2) 5
- (3) 17
- (4) 5
- (5) 6
- (6) 6

F98b

- (1) 8
- (2) 10
- (3) 25
- (4) 9
- (5) 5
- (6) 15

F99a

- (1) 13
- (2) 10
- (3) 5
- (4) 12
- (5) 7
- (6) 24

F100a

- (1) 9
- (2) 17
- (3) 3
- (4) 13
- (5) 6
- (6) 9

F99b

- (1) 10
- (2) 8
- (3) 8
- (4) 10
- (5) 24
- (6) 4

F100b

- (1) 13
- (2) 9
- (3) 12
- (4) 3
- (5) 10
- (6) 12